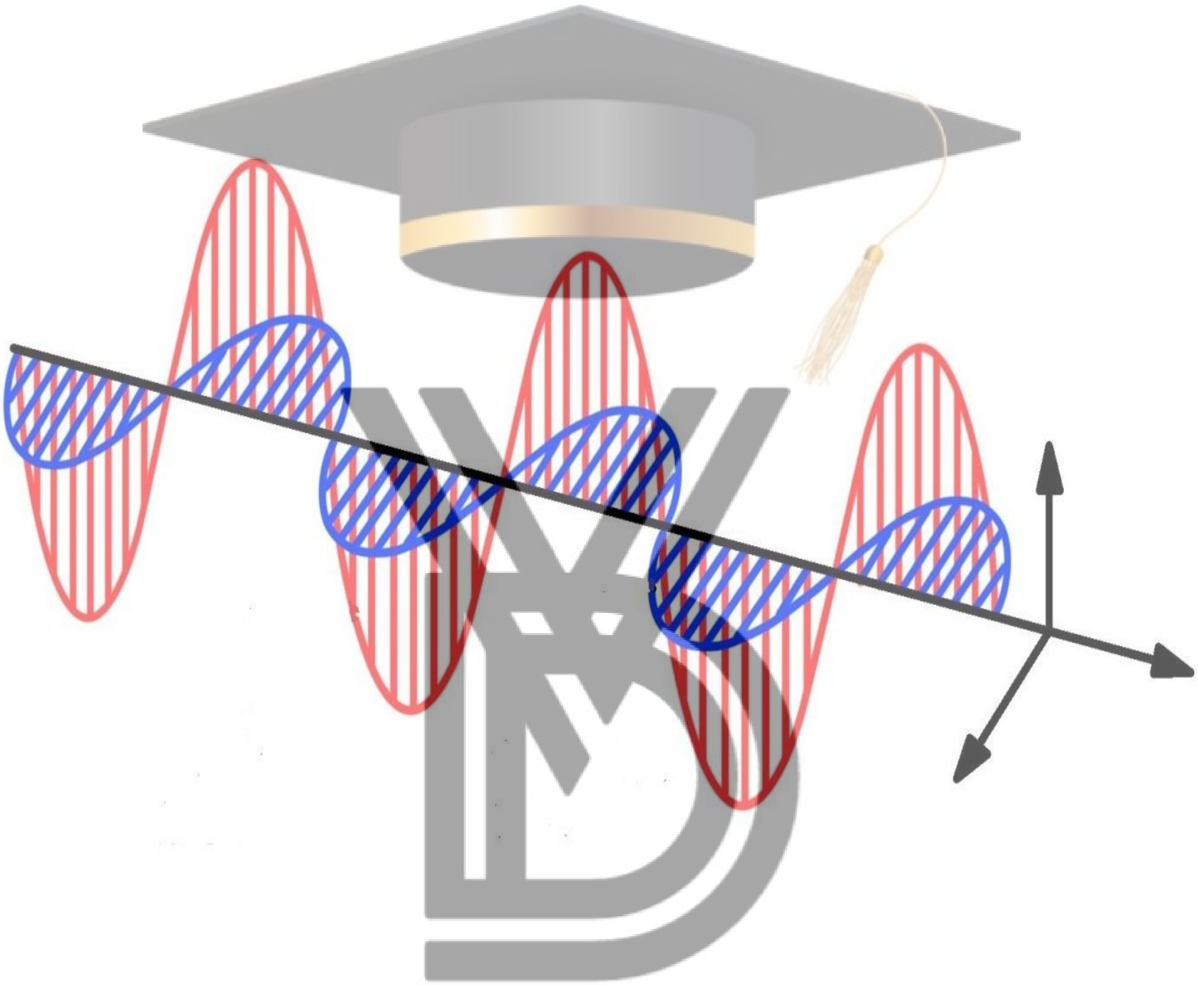
ELECTROMAGNETIC THEORY



MAHADEV S.
ASST. PROFESSOR,
DEPARTMENT OF ECE

MODULE-1

COULOMB'S LAW AND ELECTRIC FIELD INTENSITY

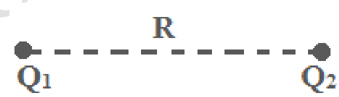
EXPERIMENTAL LAW OF COULOMB:

Coulomb invented the force exerted between two objects each having a static charge of electricity. He then stated a Coulomb's law i.e.

"The electrostatic force between two point charges Q_1 and Q_2 is directly proportional to the product of the magnitude of the charges and inversely proportional to the square of the distance between them."

The expression for the above statement is,

$$F \propto \frac{Q_1 Q_2}{R^2} \quad \text{---------1}$$



Where, Q_1 and Q_2 are charges, and

R is the distance between them.

The proportionality constant is given by k.

$$\therefore \quad F = k\frac{Q_1 Q_2}{R^2} \quad \text{---------2} \qquad \text{where } k = \frac{1}{4\pi\epsilon_0}$$

Where ϵ_0 is the permittivity of free space and the value of $\epsilon_0 = 8.854\times 10^{-12}$ F/m.

Unit of force is measured in terms of NEWTON, denoted by N.

All the units are measured in S.I. units.

Explanation:

Let us represent the Coulomb's law in vector form.

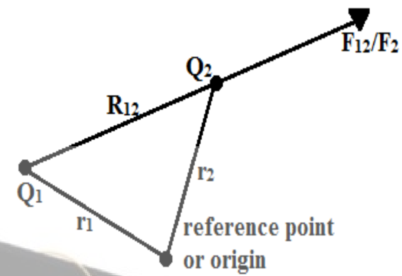
The direction of $\vec{F}_{12}$ is as shown because Q_1 and Q_2 are considered as same charge.

If, r_1 is the distance of Q_1 from origin, and

r_2 is the distance of Q_2 from origin.

From figure, force F_2 has to be calculated, then F_2 is the force on Q_2 exerted by Q_1 and from Coulomb's law,

$$F_2 = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2}\,\bar{a}_{12} \qquad \text{---------3}$$



Where, $\bar{a}_{12}$ is the unit vector in the direction of R_{12} and a_{12} is given by,

$$\bar{a}_{12} = \frac{R_{12}}{|R_{12}|} = \frac{r_2 - r_1}{|r_2 - r_1|}\,.$$

Where, R_{12} is the distance between Q_1 and Q_2 given by $r_2 - r_1$ (i.e. destination-source).

R_{12} ⇨ Vector quantity

$|R_{12}|$ ⇨ Magnitude of the vector quantity

ELECTRIC FIELD INTENSITY:

The concept of electric field intensity came to an existence due to an electric charge.

An electric charge always sets up an electric field in its surroundings, and it exerts a force upon any other charge which arrives in this field zone. Therefore, this field zone is called electric field intensity. This electric field intensity is defined as,

"The electric field intensity at any point in an electric field is the force experienced by a unit positive charge placed at that point".

If we now consider one charge fixed in position say Q_1 and move a second charge slowly around it, it experience a force in this region around the charge Q_1, which is given by,

$$F = \frac{Q_1 Q_t}{4\pi\epsilon_0 {R_t}^2} \qquad \text{--------1,} \quad \text{where } Q_t \text{ is second charge.}$$

If we consider Q_t as a unit charge, then force per unit charge is,

$$\frac{F}{Q_t} = \frac{Q_1}{4\pi\epsilon_0 {R_t}^2} \qquad \text{--------2}$$

This force per unit charge is called electric field intensity.

$$\therefore \qquad E = \frac{F}{Q_t} = \frac{Q_1}{4\pi\epsilon_0 {R_t}^2} \qquad \text{--------3}$$

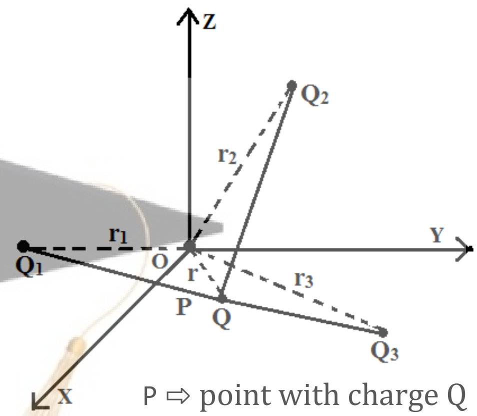
Where, E is electric field intensity and the unit is represented by Newton/Coulomb i.e. N/C or Joules/Coulombs, general representation is V/m.

ELECTRIC FIELD INTENSITY AT A POINT DUE TO MANY CHARGES:

The field at appoint when there are many charges contributing to the field can be found by employing the principle of superposition.

"If there are more than two point charges, then each will exert a force on each other, then the net force on any charge is the Principle of Superposition".

In the figure, we have considered three charges Q_1, Q_2 and Q_3 with electric field intensity E_1, E_2 and E_3.



P ⇨ point with charge Q

$\therefore \quad E = E_1 + E_2 + E_3 + \text{-------} + E_n \quad$ (for n charges)

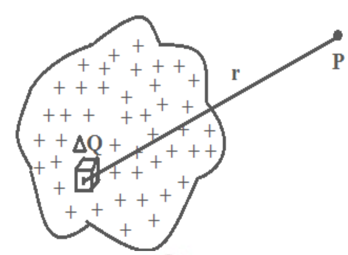
$$\Rightarrow \quad E = \frac{Q_1\, a_1}{4\pi\epsilon_0|r-r_1|^2} + \frac{Q_2\, a_2}{4\pi\epsilon_0|r-r_2|^2} + \frac{Q_3\, a_3}{4\pi\epsilon_0|r-r_3|^2} + \text{-----} + \frac{Q_n\, a_n}{4\pi\epsilon_0|r-r_n|^2}$$

$$\therefore \quad E = \sum_{m=1}^{n} \frac{Q_m\, a_m}{4\pi\epsilon_0|r-r_m|^2}.$$

FIELD DUE TO A CONTINUOUS VOLUME CHARGE DISTRIBUTION:

If the charge distribution is such that the charges are distributed continuously in a volume then it is referred to as a volume charge distribution.

ρ_v is defined as the volume charge density i.e. charge per unit volume.



$$\therefore \quad \rho_v = \frac{\Delta Q}{\Delta v} \quad \text{---------1} \quad \text{unit } c/m^3.$$

Mathematically by using limiting process for above expression,

$$\rho_v = \lim_{\Delta v \to 0} \frac{\Delta Q}{\Delta v} \quad \text{---------2}$$

But total charge within some finite volume is obtained by integrating throughout that volume.

$$\text{i.e.} \quad Q = \int_{vol} \rho_v \cdot dv \quad \text{---------3}$$

We have electric field intensity at point P due to point charge is given by,

$$\Delta E = \frac{\Delta Q}{4\pi\epsilon r^2}\, \hat{a}_r \quad \text{---------4}$$

From figure, since the charge distribution is continuous, the field at P due to the entire volume charge distribution is,

$$E = \int_{vol} \Delta E = \int_{vol} \frac{\Delta Q}{4\pi\epsilon r^2} \hat{a}_r \quad (\because \ \Delta E = \frac{\Delta Q}{4\pi\epsilon r^2} \hat{a}_r).$$

But $\Delta Q = \rho_v \cdot \Delta v$ from equation 1.

⇨ $E = \frac{1}{4\pi\epsilon_0} \int_{vol} \frac{\rho_v \cdot \Delta v}{r^2} \hat{a}_r \cdot dv$

∴ $E = \frac{\rho_v}{4\pi\epsilon_0} \int_{vol} \frac{\Delta v}{r^2} \hat{a}_r \cdot dv$

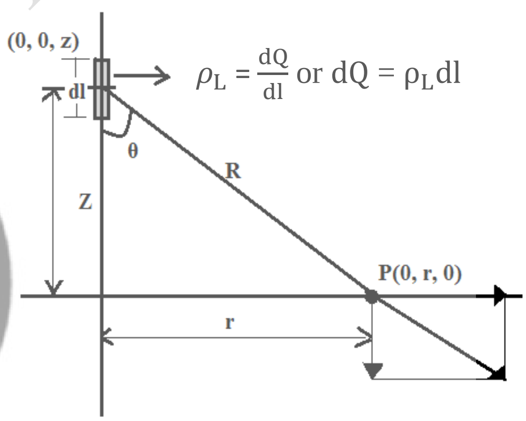
FIELD OF A LINE CHARGE:

Consider an infinitely long straight line, carrying uniform line charge having line charge density denoted by ρ_L c/m.

Let the charge on a very short segment, dl of a line be dQ. Then,

$\rho_L = \frac{dQ}{dl}$ (c/m) ---------1

⇨ $dQ = \rho_L \cdot dl.$ ---------2



Now let us calculate electric field intensity at a point due to infinitely long straight conductor, i.e. line charge varying from $-\infty$ to ∞.

Let P be the point on the y axis, at which electric field intensity is to be determined, and r be the distance of a point P from origin as shown in the figure.

Now let us calculate electric field intensity at a point P, due to entire line charge present on z-axis.

Now, let us consider a small length dl on the z-axis as shown in figure, carrying a charge dQ at point (0, 0, z). Then,

$dQ = \rho_L \cdot dl$ form equation (2).

But $dl = dz$ because of a small length.

⇨ $dQ = \rho_L \cdot dz.$ ---------3

From figure dQ has a coordinates as (0, 0, z) and point P has a coordinates of (0, r, 0).

Then the distance $R = r\,\bar{a}_y - z a_z$ and

$|R| = \sqrt{r^2 + z^2}$ {(0-0, r-0, 0-z)}

∴ Unit vector is given by, $\hat{a}_r = \frac{R}{|R|}$

⇨ $\hat{a}_r = \frac{R}{|R|} = \frac{r\,a_y - z a_z}{\sqrt{r^2+z^2}}$ ---------4

But we have electric field intensity at a point charge is given by,

$\Delta E = \frac{\Delta Q}{4\pi\epsilon_0 R^2}\,\hat{a}_R.$

Substituting the value of dQ from equation (3) and (4),

⇨ $\Delta E = \frac{\rho_L \cdot dz}{4\pi\epsilon_0 R^2}\left(\frac{r\,a_y - z a_z}{\sqrt{r^2+z^2}}\right).$

⇨ $\Delta E = \frac{\rho_L \cdot dz\,(r\,a_y - z a_z)}{4\pi\epsilon_0 (r^2+z^2)^{\frac{3}{2}}}$ --------5 $(\because R = \sqrt{r^2 + z^2})$

Important Note: since the line charge on z-axis is symmetry to y-axis then the electric field intensity produced by all charges get cancelled. Hence there are no z-components for the above equation 5.

⇨ $\Delta E = \frac{\rho_L}{4\pi\epsilon_0}\left(\frac{r\,a_y}{(r^2+z^2)^{\frac{3}{2}}}\right)\cdot dz$ --------6

The above equation gives the electric field generated by small charge dQ over the length then the net electric field intensity is given by integrating over entire length. (i.e. $-\infty$ to $+\infty$)

∴ $E = \int_{-\infty}^{+\infty} \frac{\rho_L}{4\pi\epsilon_0}\left(\frac{r\,a_y}{(r^2+z^2)^{\frac{3}{2}}}\right)\cdot dz$ --------7

From the figure, we have,

$\tan\theta = \frac{r}{z}$ or $\cot\theta = \frac{z}{r}$

⇨ $z = r\cot\theta$

Differentiating with respect to θ, we get,

⇨ $dz = -r\,\mathrm{cosec}^2\theta \cdot d\theta$

Substituting the limits,

For $z = -\infty \qquad \theta = \cot^{-1}(-\infty) = \pi$

$z = \infty \qquad \theta = \cot^{-1}(\infty) = 0$

∴ Equation (7) becomes,

$$E = \int_{\pi}^{0} \frac{\rho_L}{4\pi\epsilon_0}\left(\frac{r}{(r^2+z^2)^{\frac{3}{2}}}\right)\cdot dz \cdot a_y$$

∴ $$E = \int_{\pi}^{0} \frac{\rho_L\, r}{4\pi\epsilon_0}\left(\frac{-r\,\mathrm{cosec}^2\theta}{(r^2+r^2\cot^2\theta)^{\frac{3}{2}}}\right)\cdot a_y \cdot d\theta$$

⇨ $$E = \int_{\pi}^{0} \frac{-\rho_L}{4\pi\epsilon_0}\left(\frac{r^2\,\mathrm{cosec}^2\theta}{(r^{\cancel{2}})^{\frac{3}{\cancel{2}}}(1+\cot^2\theta)^{\frac{3}{2}}}\right)\cdot a_y \cdot d\theta$$

⇨ $$E = \frac{-\rho_L \cdot a_y}{4\pi\epsilon_0}\int_{\pi}^{0}\left(\frac{\cancel{r^2}\,\mathrm{cosec}^2\theta}{\cancel{r^3}\,(\mathrm{cosec}^{\cancel{2}}\theta)^{\frac{3}{\cancel{2}}}}\right)\cdot d\theta$$

⇨ $$E = \frac{-\rho_L \cdot a_y}{4\pi\epsilon_0 r}\int_{\pi}^{0}\left(\frac{\mathrm{cosec}^{\cancel{2}}\theta}{\mathrm{cosec}^{\cancel{3}}\theta}\right)\cdot d\theta$$

⇨ $$E = \frac{-\rho_L \cdot a_y}{4\pi\epsilon_0 r}\int_{\pi}^{0}\left(\frac{1}{\mathrm{cosec}\,\theta}\right)\cdot d\theta$$

⇨ $$E = \frac{-\rho_L \cdot a_y}{4\pi\epsilon_0 r}\int_{\pi}^{0}\sin\theta \cdot d\theta$$

⇨ $$E = \frac{-\rho_L \cdot a_y}{4\pi\epsilon_0 r}\left[-\cos\theta\right]_{\pi}^{0}$$

⇨ $$E = \frac{\rho_L \cdot a_y}{4\pi\epsilon_0 r}\left[\cos\theta\right]_{\pi}^{0}$$

⇨ $$E = \frac{\rho_L \cdot a_y}{4\pi\epsilon_0 r}\left[\cos 0 - \cos\pi\right]$$

⇨ $$E = \frac{\rho_L \cdot a_y}{4\pi\epsilon_0 r}\left[1-(-1)\right]$$

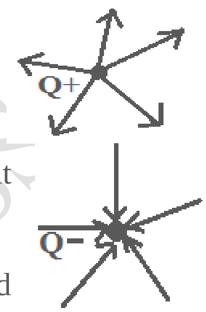
⇨ $$E = \frac{\rho_L \cdot a_y}{4\pi\epsilon_0 r}\left[2\right]$$

⇨ $E = \frac{\rho_L}{2\pi\epsilon_0 r} \cdot a_y$

∴ Above equation is electric field intensity due to infinitely long straight conductor.

ELECTRIC FLUX (Ψ):

If a unit test charge is placed near a point charge, it experiences a force. The direction of this force can be represented by the lines, coming radially outward from a positive charge. These lines are called flux lines.



Thus the electric field due to a charge can be imagined to be present around it in terms of a quantity called electric flux or displacement flux.

Michael Faraday performed the experiment on electric field and induced some rules as listed below,

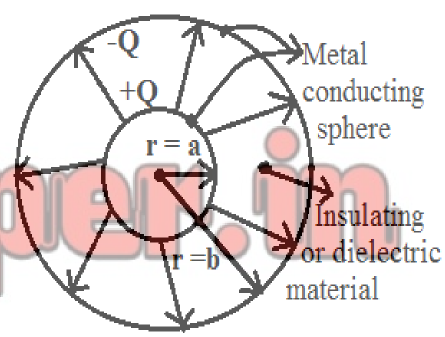
- Faraday found that the total charge on the outer sphere was equal in magnitude to the original charge placed on the inner sphere, and this was true regardless of the dielectric material separating the two spheres.
- He also showed that a large positive charge on the inner sphere induced a correspondingly larger negative charge on the outer sphere. This concept leads to a directly proportionality between the electric flux and the charge on the inner sphere.

i.e. $\Psi = Q$

∴ Electric flux is measured in terms of Coulombs.

ELECTRIC FLUX DENSITY (D):

The electric flux density is defined as total number of flux lines passing normally through the unit surface area is called electric flux density and it is represented by D.



Unit for this is Coulomb/m^2 (C/m^2).

$D = \frac{\Psi}{s}$

To explain this let us consider the following figure.

At the surface of the inner sphere, Ψ coulombs of electric flux are produced by the charge Q coulomb which is distributed uniformly over a surface having an area of $4\pi a^2$.

⇨ $D = \frac{\Psi}{4\pi a^2}$ for inner sphere.

And the direction of the flux line at that point gives the direction of D.

⇨ $D = \frac{\Psi}{4\pi a^2} a_r$ --------1 for inner sphere.

⇨ $D = \frac{\Psi}{4\pi b^2} a_r$ --------2 for outer sphere.

If we now let the inner sphere become smaller and smaller, while still retaining a charge Q, then it becomes a point charge in the limit, then the electric flux density at point r meter from the point charge is given by,

$D = \frac{Q}{4\pi r^2} a_r$ --------3 ($\because$ $\Psi = Q$).

We can find the relation between D and E, because we have

$E = \frac{Q}{4\pi\epsilon_0 r^2} a_r$ --------4

From equation (3), if we compare equation (4),

⇨ $E = \frac{D}{\epsilon_0}$ ($\because D = \frac{Q}{4\pi r^2} a_r$)

∴ $D = \epsilon_0 E$ for free space only.

From this relation we can find the flux density for volume charge distribution in free space.

Since we have, $E = \int_{vol} \frac{\rho_v \cdot dv}{4\pi\epsilon_0 R^2} a_R$

⇨ $D = \int_{vol} \frac{\rho_v \cdot dv}{4\pi R^2} a_R$

IMPORTANT FORMULAE:

1. Force between two charges "F":

$$\mathbf{F}_2 = \frac{\mathbf{Q_1 \cdot Q_2}}{4\pi\epsilon_0 R_{12}{}^2}\,\hat{\mathbf{a}}_{12} \quad \mathbf{N}$$

Where, $\epsilon_0 = 8.854\times 10^{-12}$,

R = distance between Q_1 and Q_2

$\hat{a}_{12} = \frac{R_{12}}{|R_{12}|}$, is unit vector and

Where, $R_{12} = R_2 - R_1$, vector form & $|R_{12}|$ ⇨ Magnitude.

2. Electric Field Intensity "E":

$$\mathbf{E} = \frac{\mathbf{Q_1}}{\mathbf{4\pi\epsilon r^2}}\,\boldsymbol{a}_\mathbf{r} \qquad \mathbf{V/m}$$

Where, $a_r = \frac{r}{|r|^2}$ & r ⇨ distance

3. Electric Field Intensity due to a Many Charges

$$\mathbf{E} = \sum_{\mathbf{m=1}}^{\mathbf{n}} \frac{\mathbf{Q_m}\,\boldsymbol{a}_\mathbf{m}}{\mathbf{4\pi\epsilon_0 |r - r_m|^2}}. \qquad \mathbf{V/m}$$

4. Electric Field Intensity due to Continuous Volume Charge Distribution:

$$\mathbf{E} = \frac{\boldsymbol{\rho_v}}{\mathbf{4\pi\epsilon_0}} \int_{vol} \frac{\Delta v}{\mathbf{r^2}}\,\hat{\boldsymbol{a}}_\mathbf{r} \cdot \mathbf{d}v \qquad \mathbf{V/m}$$

Where, $\rho_v = \frac{dQ}{dv}$ or $Q = \int_v \rho_v \cdot dv$

ρ_v is different for rectangular, cylindrical and spherical components

5. Field due to Line Charge:

$$\mathbf{E} = \frac{\boldsymbol{\rho_L}}{\mathbf{2\pi\epsilon_0 r}} \cdot \mathbf{a_y} \qquad \mathbf{V/m}$$

Where, $\rho_L = \frac{dQ}{dL}$ or $Q = \int \rho_L \cdot dl$

6. Electric Flux Density "D":

$$\mathbf{D} = \frac{\Psi}{s} = \frac{Q}{s} \quad \text{or} \quad \mathbf{D} = \frac{d\Psi}{ds} \qquad c/m^2$$

$$\mathbf{D} = \frac{Q}{4\pi r^2}\mathbf{a}_r \quad \text{or} \quad \mathbf{D} = \epsilon_0 \mathbf{E}$$

MODULE-2

GUASS'S LAW:

Faraday's experiments lead to the following result, i.e.

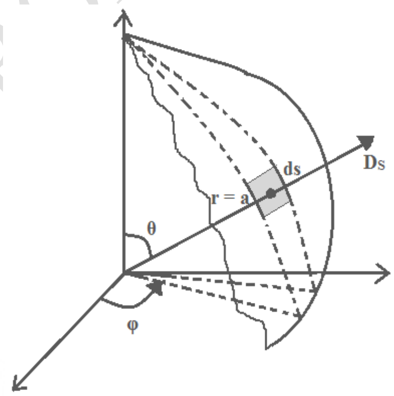
"The electric flux passing through any closed surface is equal to the total charge enclosed by that surface." The statement was called Gauss's Law.

Gauss is the person who proved the statement of

$$\Psi = Q \qquad \text{----------1}$$

Now let us prove the above statement with the help of following figure.

We have electric field intensity of a point charge, is given by,



$$E = \frac{Q}{4\pi\epsilon_0 r^2} a_r \qquad \text{--------2}$$

But we have,

$$D = \epsilon_0 E \qquad \text{--------3}$$

⇨ $$D = \frac{Q}{4\pi r^2} a_r \qquad \text{--------4}$$

Form figure, the flux density at surface having r = a, is given by,

$$D = \frac{Q}{4\pi a^2} a_r \qquad \text{--------5}$$

But for small surface ds flux density is given by,

$$D = \frac{d\Psi}{ds}$$

⇨ $$d\Psi = D \cdot ds$$

⇨ $$\Psi = \int D \cdot ds \qquad \text{--------6}$$

Figure shown is spherical surface,

⇨ Surface area of the differential element is given by,

$$ds = r^2 \sin\theta \cdot d\varphi \cdot d\theta \cdot a_r \qquad \text{--------7}$$

Substituting equation (7) and equation (5) in equation (6), we get,

$$\Psi = \oint_s D\cdot\, ds$$

$$\Psi = \oint \frac{Q}{4\pi a^2} a_r \cdot r^2 \sin\theta \cdot d\varphi \cdot d\theta \cdot a_r$$

At r = a,

$\Rightarrow$ $$\Psi = \int_{\varphi=0}^{\varphi=2\pi} \int_{\theta=0}^{\theta=\pi} \frac{Q}{4\pi \cancel{a^2}} \cdot \cancel{a^2} \sin\theta \cdot d\varphi \cdot d\theta$$

$\Rightarrow$ $$\Psi = \int_{\varphi=0}^{\varphi=2\pi} \frac{Q}{4\pi} [-\cos\theta]_0^{\pi} \cdot d\varphi$$

$\Rightarrow$ $$\Psi = \frac{Q}{4\pi} \{-[(-1) - (1)]\} \cdot [\varphi]_0^{2\pi}$$

$\Rightarrow$ $$\Psi = \frac{Q}{\cancel{4\pi}} \cancel{(2)} \cdot \cancel{2\pi}$$

$\Rightarrow$ $$\Psi = Q.$$

∴ Q coulombs of flux are passing the surface area.

We have $\Psi = \int D\cdot\, ds$

$\Rightarrow$ $Q = \oint D\cdot\, ds$

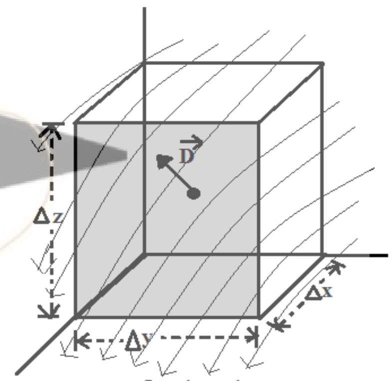
But $Q = \int_v \rho_v \cdot dv$

$\Rightarrow$ $\oint D\cdot\, ds = \int_v \rho_v \cdot dv$

$\Rightarrow$ Total electric flux through any closed surface is equal to the charge enclosed by that surface.

DIVERGENCE:

Consider a differential volume element of volume Δv as shown in figure, with the region having the electric charges. Let Δx, Δy, and Δz be the dimensions of the element as shown.



flux through a volume element

Consider a point P exactly at the center point of the volume element. Due to the field produced by the charges in the region, then the electric flux density at point P be D, where it is having a component D_x, D_y, and D_z.

The value of D at point P may be expressed in rectangular component represented as,

$$D = D_{x_0} a_x + D_{y_0} a_y + D_{z_0} a_z \qquad \text{--------1}$$

But from Gauss's law,

$$\oint_s D \cdot ds = Q. \qquad \text{--------2}$$

Closed surface integral has to be divided into six faces, i.e.

$$\oint_s D \cdot ds = \int_{front} + \int_{back} + \int_{left} + \int_{right} + \int_{top} + \int_{bottom} = Q. \qquad \text{--------3}$$

Considering any one surface, since surface area is very small D is essentially constant and is given by,

$$\int_{front} = D_{front} \cdot \Delta s_{front} \qquad \text{--------4}$$

$$\int_{front} = D_{front} \cdot \Delta y \cdot \Delta z a_x$$

Since it is in x-direction,

$$\Rightarrow \int_{front} = D_{front} \cdot \Delta y \cdot \Delta z \qquad \text{--------5}$$

The front face is at distance of $\frac{\Delta x}{2}$ from P, then,

$$D_{x\,front} = D_{x_0} + \frac{\Delta x}{2} \times \text{rate of change of } D_x \text{ with x.}$$

$$\Rightarrow \quad D_{x\,front} = D_{x_0} + \frac{\Delta x}{2} \cdot \frac{\partial D_x}{\partial x}$$

Where D_{x_0} is the value of D_x at P.

⇨ Equation (5) becomes,

⇨ $\int_{front} = \left(D_{x_0} + \frac{\Delta x}{2} \cdot \frac{\partial D_x}{\partial x}\right) \cdot \Delta y \cdot \Delta z$ --------6

Similarly for back surface the distance is $-\frac{\Delta x}{2}$ from the point P.

∴ $D_{x\,back} = -[D_{x_0} + \frac{\partial D_x}{\partial x} \cdot (-\frac{\Delta x}{2})]$

⇨ $D_{x\,back} = -[D_{x_0} - \frac{\Delta x}{2} \cdot \frac{\partial D_x}{\partial x}] = -D_{x_0} + \frac{\Delta x}{2} \cdot \frac{\partial D_x}{\partial x}$.

Negative sign is taken because the direction of flux is in opposite direction.

⇨ $\int_{back} = \left(-D_{x_0} + \frac{\Delta x}{2} \cdot \frac{\partial D_x}{\partial x}\right) \cdot \Delta y \cdot \Delta z$ --------7

Adding equations (6) and (7), we get,

$\int_{front} + \int_{back} = \frac{\partial D_x}{\partial x} \cdot \Delta x \cdot \Delta y \cdot \Delta z$ --------8

Similarly, $\int_{top} + \int_{bottom} = \frac{\partial D_z}{\partial z} \cdot \Delta x \cdot \Delta y \cdot \Delta z$ --------9

$\int_{left} + \int_{right} = \frac{\partial D_y}{\partial y} \cdot \Delta x \cdot \Delta y \cdot \Delta z$ --------10

Substituting equations (8), (9) and (10) in equation (3), we have,

$$\oint_s D \cdot ds = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}\right) \cdot \Delta x \cdot \Delta y \cdot \Delta z = Q.$$

⇨ $\left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}\right) = \frac{Q}{\Delta x \cdot \Delta y \cdot \Delta z} = \frac{Q}{\Delta v}$.

Applying the limiting factor as $\Delta v \to 0$, we get,

⇨ $\left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}\right) = \lim_{\Delta v \to 0} \frac{Q}{\Delta v} = \lim_{\Delta v \to 0} \frac{\oint_s D \cdot ds}{\Delta v} = \rho_v$ ·

i.e. $\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \lim_{\Delta v \to 0} \frac{Q}{\Delta v}$. --------11

∴ Above equation is called Divergence.

∴ The divergence of the vector flux density D is the outflow of flux from a small closed surface per unit volume as the volume shrinks to zero.

⇨ $\nabla \cdot D = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$, for rectangular component.

⇨ $\nabla \cdot D = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho \cdot D_\rho) + \frac{1}{\rho}\frac{\partial D_\varphi}{\partial \varphi} + \frac{\partial D_z}{\partial z}$, for cylindrical component.

⇨ $\nabla \cdot D = \frac{1}{r^2}\frac{\partial}{\partial x}(r^2 \cdot D_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta \cdot D_\theta) + \frac{1}{r\sin\theta}\frac{\partial D_\varphi}{\partial \varphi}$,

for spherical component.

From Divergence we have,

$$\nabla \cdot D = \frac{\Delta Q}{\Delta v}$$

But $\rho_v = \frac{\Delta Q}{\Delta v}$

∴ $\nabla \cdot D = \rho_v$

This is the Maxwell's first equation and it is also called as Gauss's Law in point form or differential form. Statement of Gauss's Law in point form,

"The divergence of electric flux density in a medium at a point is equal to the charge per unit volume at the same point."

From Gauss's Law, we have,

$$Q = \oint_s D \cdot ds$$

But continuous charge distribution in volume is given by,

$$Q = \int_v \rho_v \cdot dv$$

From the definition of divergence we have, $\rho_v = \nabla \cdot D$.

⇨ $Q = \oint_s D \cdot ds = \int_v \rho_v \cdot dv = \int_v \nabla \cdot D \cdot dv$

⇨ $\oint_s D \cdot ds = \oint_v \nabla \cdot D \cdot dv$

The above equation is called divergence theorem.

The divergence theorem states that,

"The integral of normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field throughout the volume enclosed by the closed surface."

ENERGY AND POTENTIAL

ENERGY EXPANDED IN MOVING A POINT CHARGE IN AN ELECTRIC FIELD:

The electric field intensity was defined as the force on unit test charge that point at which we wish to find the value of this vector field. In this electric field we try to move the charge against the electric field, for this we have to exert a force equal and opposite to that exerted by the field, due to this it exert a negative charge.

Suppose we wish to move a charge Q a distance dl in an electric field E.

∴ The force on Q arising from the electric field is,

$$F_E = Q \cdot E$$

Or in vector form,

$$F_E = F \cdot a_L$$

$$F_E = Q \cdot E \cdot a_L$$

Since the force which we must apply is equal and opposite to the force and distance.

$$\therefore \quad F_{apply} = -Q \cdot E \cdot \hat{a}_L$$

i.e. Differential work done by external source moving through small distance dl is given as,

$$dW = -Q \cdot E \cdot dL$$

∴ The total work done in moving a point charge from infinite to finite distance is given by,

$$W = -Q \int_{infinite}^{finite} E \cdot dL$$

THE LINE INTIGRAL:

Now let us calculate the work done on a charge moving from infinite distance is given by,

$$W = -Q \int_{\text{infinite}}^{\text{finite}} E \cdot dL$$

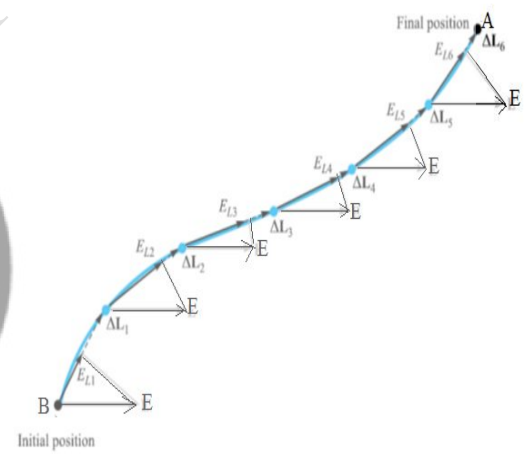
Let us calculate the work done for a line integral for the figure shown.

Now let us calculate a work done on a path B to A over a uniform electric field as shown in the figure.

The path is divided into six segment $\Delta L_1, \Delta L_2, \Delta L_3, \Delta L_4, \Delta L_5, \Delta L_6$ and a component of E along each segment is denoted by $E_{L1}, E_{L2}, \ldots\ldots E_{L6}$.

$\therefore$ The work involved in moving a charge Q from B to A is given by,

$$W = -Q\,[E_{L1} \cdot \Delta L_1 + E_{L2} \cdot \Delta L_2 + \cdots\cdots + E_{L6} \cdot \Delta L_6]$$



Using vector notation, we get,

$$W = -Q\,[E_1 \cdot \Delta L_1 + E_2 \cdot \Delta L_2 + \cdots\cdots + E_6 \cdot \Delta L_6]$$

But we have assumed uniform field,

$$\Rightarrow \quad E = E_1 = E_2 = \cdots\cdots = E_6$$

$$\therefore \quad W = -Q \cdot E\,[\Delta L_1 + \Delta L_2 + \cdots\cdots + \Delta L_6]$$

But,

$$\Delta L_1 + \Delta L_2 + \cdots\cdots + \Delta L_6 = L_{BA}$$

$$\therefore \quad W = -Q \cdot E\, L_{BA}$$

But we have summation of small value of ΔL is nothing but the integration,

$$\therefore \quad W = -Q \cdot E \int_B^A dL$$

From this expression it is clear that work involved in moving the charge depends on Q, E and L_{BA}, a vector drawn from the initial to the final point of the path chosen.

Note: dL will have following value.

$$dL = dx \cdot a_x + dy \cdot a_y + dz \cdot a_z \qquad \textbf{(Rectangular)}$$

$$dL = d\rho \cdot a_\rho + \rho d\varphi \cdot a_\varphi + dz \cdot a_z \qquad \textbf{(cylindrical)}$$

$$dL = dr \cdot a_r + r d\theta \cdot a_\theta + r \sin\theta \, d\varphi \cdot a_\varphi \qquad \textbf{(spherical)}$$

But electric field intensity of line integral is,

$$E = E_r a_r = \frac{\rho_L}{2\pi\epsilon_0 r} \cdot a_r$$

$$\therefore \text{ Work done} = -Q \int_{\text{initial}}^{\text{final}} \frac{\rho_L}{2\pi\epsilon_0 r} \cdot a_r \cdot a_r \cdot dr$$

$$= -Q \frac{\rho_L}{2\pi\epsilon_0} \int_b^a \frac{1}{r} \cdot dr$$

$$= -\frac{Q\rho_L}{2\pi\epsilon_0} [\log r]_b^a$$

$$= -\frac{Q\rho_L}{2\pi\epsilon_0} \log[a] - \log[b]$$

$$W = \frac{Q\rho_L}{2\pi\epsilon_0} \log\left(\frac{b}{a}\right)$$

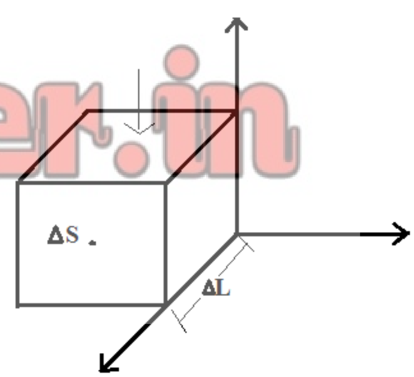
CURRENT AND CONDUCTORS:

CURRENT AND CURRENT DENSITY:

Electric charges in motion constitute a current. The unit of current is Ampere (A), it is also defined as a rate of movement of charge passing a given reference point of one coulomb per second, current symbol is given by,

$$\therefore \quad I = \frac{dQ}{dt}.$$

Now let us define the current density measured in A/m^2, current density is a vector represented by J.



The incremental current crossing an incremental surface Δs normal to the current density is,

$$\Delta I = J_N \cdot \Delta s$$

Now, if current density is not perpendicular to the surface, then $\Delta Q = \rho_v \cdot \Delta V$

$$\Delta I = J \cdot \Delta s$$

∴ Total current is obtained by integrating.

$$I = \int_s J \cdot ds$$

Fig a

Current density may be related to velocity of volume charge density at a point, as shown in fig.

We know that, $\Delta V_x = \frac{\Delta x}{\Delta t}$

i.e. rate of change of distance with respect to time.

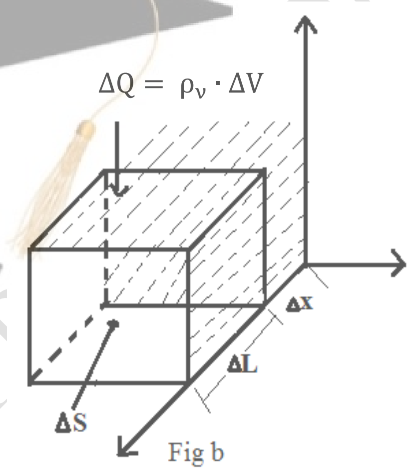


Fig b

Now, from figure it is clear that the element has moved a distance Δx, as shown in figure b.

$$\Delta Q = \rho_\nu \cdot \Delta V = \rho_\nu \cdot \Delta s \cdot \Delta L$$

Now since it has travelled a distance dx,

⇨ $\Delta Q = \rho_\nu \cdot \Delta s \cdot \Delta x$

through a reference plane perpendicular to the direction of motion in a time increment ∇t.

∴ $\Delta I = \frac{\Delta Q}{\Delta t} = \rho_\nu \cdot \Delta s \cdot \frac{\Delta x}{\Delta t}$

$$\Delta I = \rho_\nu \cdot \Delta s \cdot V_x$$

V_x represent x component of the velocity.

⇨ Current density, $J_x = \rho_\nu \cdot V_x$

In general, $J = \rho_\nu \cdot V$

∴ The above equation clearly shows that, charge in motion constitutes a current, we call this type of current as convection current J or $\rho_\nu \cdot V$ is the convection current density.

CONTINUITY OF CURRENT:

The continuity equation of current is based on the principle of conservation of charge.

The principle states that, "The charge can be neither created nor be destroyed".

But we have $I = \oint_s J \cdot ds$, total current passing through the surface s.

But we also know that current is nothing but the flow of positive charges. Hence the current I, is obtained due to the outward flow of positive from the closed surface.

According to the principle of conservation of charge, there must be decrease of an equal amount of positive charge inside the closed surface.

Hence the outward rate of flow of positive charge gets balanced by the rate of decrease of charge inside the closed surface.

Let Q_i = charge within the closed surface, then

$-\frac{dQ_i}{dt}$ = rate of decrease of charges inside the closed surface.

Negative sign indicates the decrease of charges.

⇨ $I = \oint_s J \cdot ds = -\frac{dQ_i}{dt}$

This is the integral form of the continuity equation of the current.

From the above equation we can derive the point form of continuity equation with the help of divergence theorem.

$$\oint_s J \cdot ds = \oint_v \nabla \cdot J \cdot dv$$

$$-\frac{dQ_i}{dt} = \oint_v (\nabla \cdot J) \cdot dv$$

But we have, $Q_i = \oint_v \rho_v \cdot dv$

⇨ $-\frac{d}{dt}\oint_v \rho_v \cdot dv = \oint_v (\nabla \cdot J) \cdot dv$

If we keep the surface constant, then derivative becomes partial derivative.

⇨ $\oint_v (\nabla \cdot J) \cdot dv = \int_v -\frac{\partial \rho_v}{\partial t} \cdot dv$

Since the expression is true for any volume, then it is true for small incremental volume and is given by,

$$\nabla \cdot J \cdot \cancel{dv} = -\frac{\partial \rho_v}{\partial t} \cdot \cancel{dv}$$

⇨ $\nabla \cdot J = -\frac{\partial \rho_v}{\partial t}$ The above mentioned equation is the point form or differential form of the continuity equation of the current.

MODULE – 3

POISSON'S EQUATIONS AND LAPLACE EQUATIONS

Till now we have learned few methods of finding the electric field intensity, given a known charge distribution or a potential field.

We have learned more about the close relationship between the potentials and electric fields.

In this unit we are going to solve Laplace and Poisson's equations. These equations enable us to find potential field with in the regions, bounded by known potential or charge density.

Laplace and Poisson's equations, when compared to our other methods, are probably the most widely used.

Derivation of Poisson's and Laplace Equation

From the Gauss's Law we have

$$\nabla \cdot D = \rho_v \quad \text{-------------} (1)$$

$$\text{But } D = \varepsilon E \quad \text{-------------} (2)$$

$$\nabla \cdot \varepsilon E = \rho_v$$

$$\varepsilon \cdot \nabla \cdot E = \rho_v$$

$$\nabla \cdot E = \frac{\rho_v}{\varepsilon} \quad \text{-------------} (3)$$

We also have $E = -\nabla \cdot V$

$$\nabla \cdot (-\nabla \cdot V) = \frac{\rho_v}{\varepsilon}$$

$$\boldsymbol{\nabla^2 V = -\frac{\rho_v}{\varepsilon}} \quad \text{-------------} (4)$$

Equation (4) is called Poisson's Equation, which is applicable to homogeneous medium.

If $\rho_v = 0$, indicating zero volume charge density, but allowing point charges, line charges and surface charge density to exist at singular location, then

$$\boldsymbol{\nabla^2 V = 0} \quad \text{-------------} (5)$$

The above equation (5) is called Laplace Equation.

Laplace equation in three different co-ordinates:

- Rectangular co-ordinates/Cartesian co-ordinates system ;

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

- Cylindrical co-ordinates system;

$$\nabla^2 V = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial \emptyset^2} + \frac{\partial^2 V}{\partial z^2}$$

- Spherical co-ordinates system;

$$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2(\sin\theta)^2}\frac{\partial^2 V}{\partial \emptyset^2}$$

THE STADY MAGNETIC FIELD

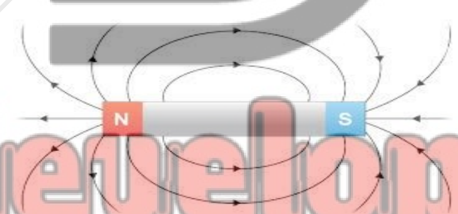
At this point we have studied a lot of concept of electric field. Now we study the effect of the magnetic field and its properties.

Magnetic field and its properties:

To understand the properties of magnetic field we take simple magnet as an example which has North pole and South pole.

Therefore the region around a magnet within which the influence of the magnet can be experienced is called Magnetic field.

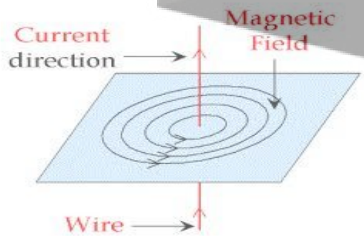
These fields are represented by imaginary lines around the magnet, which are called magnetic lines of forces, or Magnetic flux lines and the direction of these lines are always from North to South pole, which means magnetic line of forces exists in closed loops only as shown in figure.

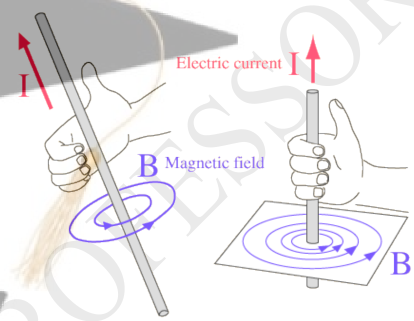


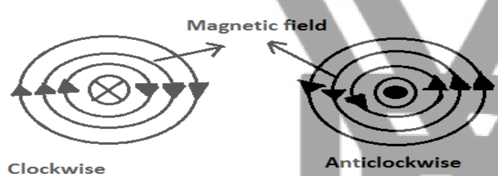
Magnetic field due to current carrying conductor:

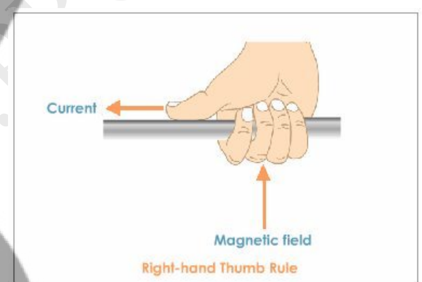
A right hand thumb rule is used to determine the direction of magnetic field around a conductor carrying a direct current.

“It states that, hold the current carrying conductor in the right hand such that the thumb pointing in the direction of current and parallel to the conductor, then curled fingers points the direction of magnetic lines of flux around it, as shown in figure.”

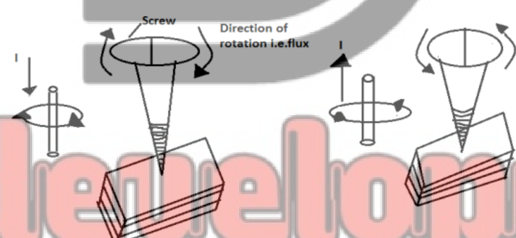








Another method of identifying the direction of magnetic flux around a conductor is right hand screw rule; it states that the direction of magnetic field is given by the direction in which the screw must be turned, as the screw moves in the direction of current.



Magnetic flux lines always forms a closed loop and is denoted by Φ. It is measured in Weber Wb.

Magnetic field Intensity:

The magnetic field intensity at any point in the magnetic field is defined as force experienced by a unit north pole of 1Wb strength, when placed at that point and is measured in Newton/Weber (N/Wb) or Ampere/meter (A/m). It is denoted by **H.**

Magnetic flux density:

Magnetic flux crossing a unit area in a plane at right angles to the direction of flux is called magnetic flux density. It is denoted as B and is a vector quantity. It is measured in Weber/m^2 or it is also called Tesla.

Relation between B and H:

$B = \mu H = \mu_0 \mu_r H$

For free space $\mu_r = 1$

$B = \mu_0 H$ and $\mu_0 = 4\pi \times 10^{-7}$

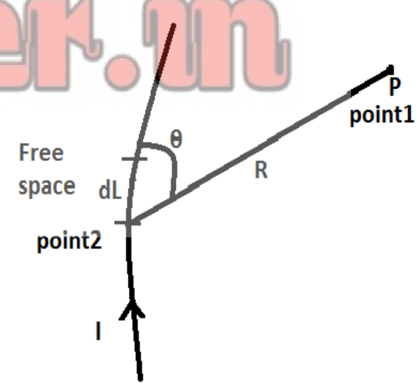
$\mu \Rightarrow$ permeability.

Biot Savart Law:

We assume a current I is flowing in a conductor as shown in figure. We have to calculate the magnitude of the magnetic field intensity, to calculate this we apply BIOT-SAVART LAW at any point P due to differential element length dl.

The magnitude of the magnetic field intensity produced by the differential element length is proportional to the product of the current, the magnitude of the differential length, and sine of angle made between the filament and a line connecting the filament to the point P and inversely proportional to the square of the distant from the differential element to the point P. The direction of the magnetic field intensity is normal to the plane containing the differential element and the line joining the filament to the point P.

The BIOT-SAVART LAW mathematically represented by



$$dH \propto \frac{Idl \sin\theta}{R^2}$$

$$\Rightarrow \quad dH = \frac{kIdl \sin\theta}{R^2}$$

Where $k = \frac{1}{4\pi}$

$$\Rightarrow \quad dH = \frac{Idl \sin\theta}{4\pi R^2}$$

The vector notation of above expression is given by considering the direction along a_R.

$$\Rightarrow \quad dH = \frac{Idla_R \sin\theta}{4\pi R^2}$$

But we have from cross product

$$dla_R \sin\theta = dl \times a_R$$

$$\Rightarrow \quad dH = \frac{I\, dl \times a_R}{4\pi R^2}$$

The integral form of BIOT-SAVART LAW is given by

$$H = \oint \frac{I\, dl \times a_R}{4\pi R^2} \qquad \text{A/m}$$

Where $\quad a_R = \frac{R}{|R|}$

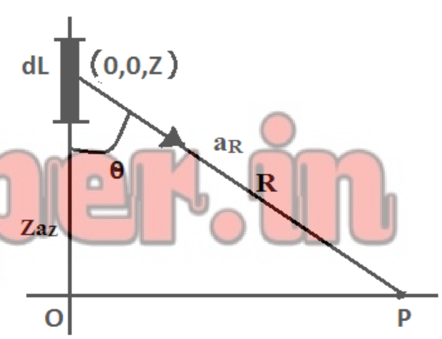
BIOT-SAVART LAW for infinitely long straight conductor or filament:

Let us consider an infinitely long straight current carrying conductor, along z-axis.

To calculate the magnetic field intensity at point P due to the current carrying conductor, let us consider dl i.e. differential element in z-axis at a distant of z.

We have from BIOT-SAVART LAW,

$$H = \oint \frac{I\, dl \times a_R}{4\pi R^2}$$



But $a_R = \frac{R}{|R|}$

$R = \rho a_\rho - z a_z$

$$\Rightarrow \quad |R| = \sqrt{\rho^2 + z^2}$$

$$\therefore \quad a_R = \frac{\rho a_\rho - z a_z}{\sqrt{\rho^2 + z^2}}$$

$$\therefore \quad H = \oint \frac{I\, dl \times R}{4\pi R^3}$$

$$\Rightarrow \quad dl\times R = \begin{vmatrix} a_\rho & a_\varphi & a_z \\ 0 & 0 & dz \\ \rho & 0 & -z \end{vmatrix}$$

$$\Rightarrow \quad dl\times R = a_\rho(0-0) - a_\varphi(0-\rho dz) + a_z(0-0)$$

$$\Rightarrow \quad dl\times R = \rho dz\, a_\varphi$$

Substitute in H expression above for infinite length,

$$H = \int_{-\infty}^{\infty} \frac{I\, dl \times R}{4\pi R^3}$$

$$\Rightarrow \quad H = \int_{-\infty}^{\infty} \frac{I\, \rho\, dz a_\varphi}{4\pi(\rho^2+z^2)^{\frac{3}{2}}}$$

$$\Rightarrow \quad H = \frac{I\, \rho\, a_\varphi}{4\pi} \int_{-\infty}^{\infty} \frac{dz}{(\rho^2+z^2)^{\frac{3}{2}}}$$

From figure, $Z = \rho \cot\theta$

$dZ = -\rho \text{cosec}^2\theta \cdot d\theta$

If $Z = -\infty \qquad \theta \Rightarrow \pi$

$Z = \infty \qquad \theta \Rightarrow 0$

$$\Rightarrow \quad H = \frac{I\, \rho\, a_\varphi}{4\pi} \int_{\pi}^{0} - \frac{\rho\, cosec^2\theta}{(\rho^2+\rho^2 cot^2\theta)^{\frac{3}{2}}}\, d\theta$$

$$\Rightarrow \quad H = -\frac{I\, \rho^2 a_\varphi}{4\pi} \int_{\pi}^{0} \frac{cosec^2\theta}{(\rho^2)^{\frac{3}{2}}(1+cot^2\theta)^{\frac{3}{2}}}\, d\theta$$

$$\Rightarrow \quad H = -\frac{I\, a_\varphi}{4\pi\rho} \int_{\pi}^{0} \frac{cosec^2\theta}{(cosec^2\theta)^{\frac{3}{2}}}\, d\theta$$

$$\Rightarrow \quad H = -\frac{I\, a_\varphi}{4\pi\rho} \int_{\pi}^{0} \frac{1}{cosec\theta}\, d\theta$$

$$\Rightarrow \quad H = -\frac{I\, a_\varphi}{4\pi\rho} \int_{\pi}^{0} \sin\theta\, d\theta$$

$$\Rightarrow \quad H = -\frac{I\, a_\varphi}{4\pi\rho} \left[-cos\theta\right]_{\pi}^{0}$$

⇨ $H = \frac{I\, a_\varphi}{4\pi\rho}[\cos 0 - \cos\pi]$

⇨ $H = \frac{I\, a_\varphi}{4\pi\rho}[1-(-1)]$

⇨ $H = \frac{I\, a_\varphi}{4\pi\rho} \times 2$

⇨ $\mathbf{H = \frac{I\, a_\varphi}{2\pi\rho}}$ **A/m**

Similarly, $B = \mu H$

⇨ $\mathbf{B = \frac{I\,\mu a_\varphi}{2\pi\rho}}$ **Wb/m²**

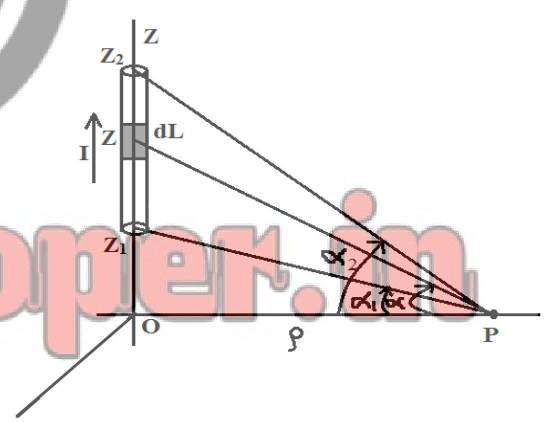
BIOT-SAVART LAW for finite current carrying conductor:

To calculate magnetic field intensity due to a finite current carrying conductor, we consider the following figure, for the length of Z_1 to Z_2.

To find H, let us consider dl from the finite length Z_1 to Z_2 having length of Z from origin and which makes angle $\propto$ with point P by connecting the point.

Similarly we have $\propto_1$ and $\propto_2$ are the angle made due to point Z_1 and Z_2 respectively with point P.

We have from BIOT-SAVART LAW,



$$H = \oint \frac{I\, dl \times a_R}{4\pi R^2}$$

But $a_R = \frac{R}{|R|}$

$R = \rho a_\rho - z a_z$

⇨ $|R| = \sqrt{\rho^2 + z^2}$

∴ $a_R = \frac{\rho a_\rho - z a_z}{\sqrt{\rho^2+z^2}}$

∴ $H = \oint \frac{I\, dl \times R}{4\pi R^3}$

⇨ $dl\times R = \begin{vmatrix} a_\rho & a_\varphi & a_z \\ 0 & 0 & dz \\ \rho & 0 & -z \end{vmatrix}$

⇨ $dl\times R = a_\rho(0-0) - a_\varphi(0-\rho dz) + a_z(0-0)$

⇨ $dl\times R = \rho dz\, a_\varphi$

Substitute in H expression above for infinite length,

$$H = \int_{-\infty}^{\infty} \frac{I\, dl\ x\ R}{4\pi R^3}$$

⇨ $H = \int_{-\infty}^{\infty} \frac{I\,\rho\, dz a_\varphi}{4\pi(\rho^2+z^2)^{\frac{3}{2}}}$

⇨ $H = \frac{I\,\rho\, a_\varphi}{4\pi} \int_{-\infty}^{\infty} \frac{dz}{(\rho^2+z^2)^{\frac{3}{2}}}$

From figure, we can select $Z_1 = \rho \tan \propto_1$ ⇨ $\propto_1 = \tan^{-1}\left(\frac{Z1}{\rho}\right)$

$Z_2 = \rho \tan \propto_2$ ⇨ $\propto_2 = \tan^{-1}\left(\frac{Z2}{\rho}\right)$

$Z = \rho \tan \alpha$

$dz = \rho sec^2\alpha\, d\alpha$

Substituting in the above equation of H, we get,

$$H = \frac{I\,\rho\, a_\varphi}{4\pi} \int_{\alpha_1}^{\alpha_2} \frac{\rho sec^2\alpha\, .\, d\alpha}{(\rho^2+\rho^2 tan^2\alpha)^{\frac{3}{2}}}$$

⇨ $H = \frac{I\,\rho^2\, a_\varphi}{4\pi} \int_{\alpha_1}^{\alpha_2} \frac{sec^2\alpha\, .\, d\alpha}{(\rho^2)^{\frac{3}{2}}(1+tan^2\alpha)^{\frac{3}{2}}}$

⇨ $H = \frac{I\,\rho^2\, a_\varphi}{4\pi\rho^3} \int_{\alpha_1}^{\alpha_2} \frac{sec^2\alpha\, .\, d\alpha}{(sec^2\alpha\,)^{\frac{3}{2}}}$

⇨ $H = \frac{I\, a_\varphi}{4\pi\rho} \int_{\alpha_1}^{\alpha_2} \frac{sec^2\alpha\, .\, d\alpha}{sec^3\alpha}$

⇨ $H = \frac{I\, a_\varphi}{4\pi\rho} \int_{\alpha_1}^{\alpha_2} \frac{1\, d\alpha}{\sec\alpha}$

⇨ $H = \frac{I\, a_\varphi}{4\pi\rho} \int_{\alpha_1}^{\alpha_2} \cos\alpha\ \ d\alpha$

⇨ $H = \frac{I\, a_\varphi}{4\pi\rho}\ [\sin\alpha]_{\alpha_1}^{\alpha_2}$

$$\mathbf{H} = \frac{\mathbf{I}\, \boldsymbol{a_\varphi}}{\mathbf{4\pi\rho}} \left[\sin(\alpha_2) - \sin(\alpha_1)\right] \qquad \mathbf{A/m}$$

But B = μH

$$\Rightarrow \mathbf{B} = \frac{\mu\, \mathbf{I}\, \boldsymbol{a_\varphi}}{\mathbf{4\pi\rho}} \left[\sin(\alpha_2) - \sin(\alpha_1)\right] \qquad \mathbf{wb/m^2}$$

MAGNETIC FLUX AND FLUX DENSITY:

Gauss's law in differential form for magnetic fields:

We have $B = \mu_0 H$, where $\mu_0 \Rightarrow$ free space permeability.

We also have $B = \frac{d\varphi}{ds}$ Wb/m^2

$d\varphi = B \cdot ds$ for small surface

$\Rightarrow \varphi = \oint_s B \cdot ds$

i.e. Magnetic flux density is defined as flux in weber passing through unit area in a plane at right angle to the direction of flux.

If the flux passing through the unit area is not exactly at right angles to the plane crossing the area but making some angle with plane then the flux φ crossing the area is given by,

$\varphi = \oint_s B \cdot ds$

$\varphi \Rightarrow$ magnetic flux density.

$B \Rightarrow$ magnetic flux density Wb/m^2.

$ds \Rightarrow$ flux passing through the area.

Since we know that magnetic flux lines always exits in closed loop, thus for a closed surface the number of magnetic flux lines entering must be equal to the number of flux lines leaving the surface, due to this no magnetic flux exist in a closed surface.

Hence $\varphi = \oint_s B \cdot ds = 0$

But from divergence theorem,

$\oint_s B \cdot ds = \oint_v \nabla \cdot B \cdot dv$

Since $dv \neq 0$,

$$\Rightarrow \nabla \cdot B = 0$$

Therefore, the divergence of magnetic flux density is always zero.

This is called Gauss's law in differential form for magnetic fields.

This is another Maxwell's equation.

Ampere's Circuital Law:

The line integral of magnetic field intensity H around a closed path is exactly equal to the direct current enclosed by that path.

$$\oint H \cdot dl = I \qquad \text{-----------}1$$

But we have already studied the magnetic field intensity H for infinitely long current carrying conductor is given by,

$$H = \frac{I}{2\pi\rho} a_{\varphi} \qquad \text{-----------}2$$

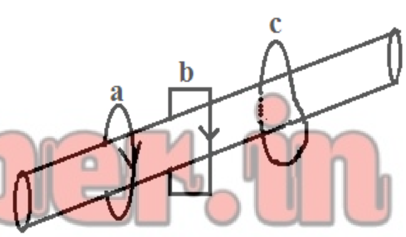
And $\qquad dl = \rho \cdot d\varphi \cdot a_{\varphi} \qquad$ -----------3

i.e. If we assume closed circular path radius ρ then dl is given by equation (3).

$$H \cdot dl = \frac{\rho \cdot d\varphi \cdot a_{\varphi} \cdot a_{\varphi} \cdot I}{2\pi\rho}$$

$$H \cdot dl = \frac{I}{2\pi\rho} d\varphi \qquad (a_{\varphi} \cdot a_{\varphi} = 1)$$

⇨ Taking Integration both side, we get



$$\oint H \cdot dl = \oint \frac{I}{2\pi\rho} d\varphi$$

$$= \int_0^{2\pi} \frac{I}{2\pi\rho} d\varphi$$

$$= \frac{I}{2\pi} [\varphi]_0^{2\pi}$$

$$= \frac{I}{2\pi} [2\pi - 0]$$

$$= I$$

⇨ $\oint H \cdot dl = I$ This proves Ampere's Circuital law.

A conductor has total current I. The line integral of H about the closed paths a and b is equal to I, and the line integral around the path c is less than I. Since the entire current is not enclosed by the path.

Area of circle = πR^2.

Perimeter of circle = $2\pi R$.

Current/unit area of cross section = $\frac{I}{\pi R^2}$.

Current enclosed by the path = $\frac{\text{current}}{\text{unit area}} \times$ area of element.

$$= \frac{I}{\pi R^2} \times \pi r^2 = I\,\frac{r^2}{R^2}.$$

Expression for H at any point due to a co-axial cable:

This is the application of Ampere's Circuital Law, consider infinitely long co-axial transmission line carrying a uniformly distributed total current I in the center conductor and –I in the outer conductor, as shown in the figure.

Symmetry shows that H is not a function of φ or Z.

Further, μ_ρ ⇨ component at $\varphi = 0$ produced by a filament located at $\rho = \rho_1$ and $\varphi = \varphi_1$, is cancelled by H_ρ component produced by $\rho = \rho_1$ and $\varphi = -\varphi_1$, as shown in fig b.

⇨ H_φ varies only with respect to ρ.

a) Consider a circular path of radius ρ such that ρ<a, and apply Ampere's Circuital Law to the closed path.

$$\oint H\cdot dl = I$$

Let us assume that the solid conductor is made up of infinitely large number of filament conductors, each carrying a current in z-direction. So that φ component of H is produced

i.e. $H = H_\varphi a_\varphi$ and $dl = \rho \cdot d\varphi \cdot a_\varphi$

⇨ $\int_0^{2\pi} \rho \cdot H_\varphi \cdot d\varphi = 2\pi\rho H_\varphi$

But we have current enclosed by the circular path = RHS = $I\,\frac{\rho^2}{a^2}$

Equating, we get

⇨ $2\pi\rho H_\varphi = I\,\frac{\rho^2}{a^2}$

⇨ $H_\varphi = \frac{\rho I}{2\pi a^2}$

b) If ρ>c ⇨ $H_\varphi = 0$ (no current is enclosed $I_{enc} = I - I = 0$)

c) For radius ρ such that $a < \rho < b$ acts as current filament.

⇨ $H_\varphi = \frac{I}{2\pi\rho}\,a_\varphi$ $(a < \rho < b)$

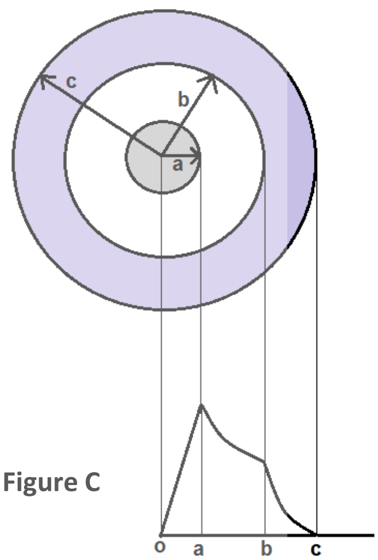


Figure C

d) b < ρ < c ⇨ Since the path lies within the outer conductor, we have

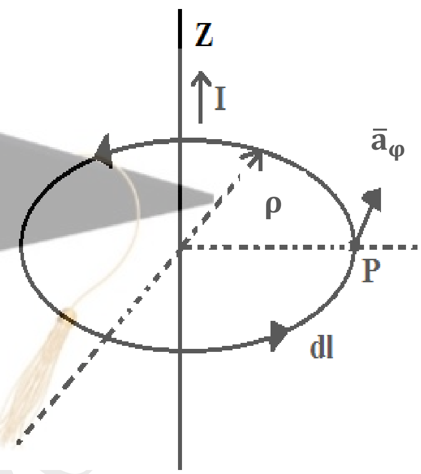
⇨ $2\pi\rho H_\varphi = I - I\left(\frac{\rho^2 - b^2}{c^2 - b^2}\right)$

⇨ $2\pi\rho \cdot H_\varphi = I\left(\frac{c^2 - \rho^2}{c^2 - b^2}\right)$

⇨ $H_\varphi = \frac{I}{2\pi\rho}\left(\frac{c^2 - \rho^2}{c^2 - b^2}\right)$.

H due to infinitely long straight conductor by using Ampere's Circuital Law:

Consider an infinitely long straight conductor placed along z-axis, carrying a direct current I as shown in the figure.



Applying the Ampere's Circuital Law we find the value of H from the figure where radius of the path is ρ and point P is the perpendicular distance ρ from the conductor, on the closed path.

Since we know that if the current is travelling along z-axis then the direction is always tangential to the closed path i.e. along a_φ.

∴ $H = H_\varphi a_\varphi$

If we consider elementary length dl at point P and long straight conductor acts as a cylinder.

⇨ $dl = \rho \cdot d\varphi \cdot a_\varphi$

∴ $H \cdot dl = H_\varphi a_\varphi \rho \cdot d\varphi \cdot$

⇨ $H \cdot dl = H_\varphi \rho d\varphi$

∴ $\oint H \cdot dl = \int_0^{2\pi} H_\varphi \rho d\varphi = I$ (∵ of closed path)

φ ⇨ 0 to 2π

⇨ $H_\varphi \rho \int_0^{2\pi} d\varphi = I$

⇨ $H_\varphi \rho [\varphi]_0^{2\pi} = I$

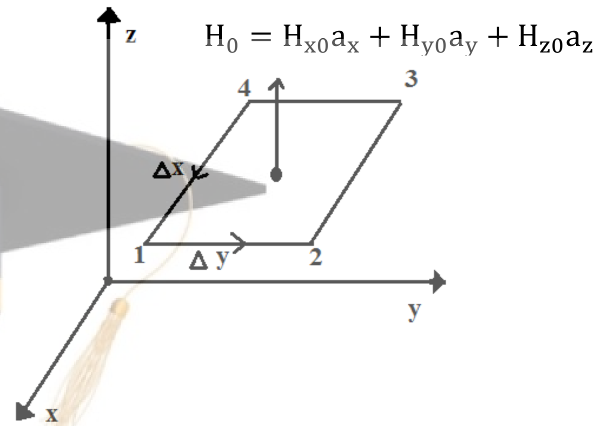
⇨ $H_\varphi \rho [2\pi - 0] = I$

⇨ $H_\varphi \rho 2\pi = I$

⇨ $H_\varphi = \frac{I}{2\pi\rho}$ A/m.

CURL:

We now apply the Ampere's Circuital Law to the perimeter of a differential surface element and discuss the third and last of the spherical derivatives of vector analysis, the curl.



Our immediate objective is to obtain the point form of Ampere's Circuital Law.

Again we shall choose rectangular co-ordinates, and an incremental closed path of side Δx and Δy is selected as shown in figure. We assume that some current, as yet unspecified produces a reference value for H at the center of this small rectangle, given by

$$H_0 = H_{x0}a_x + H_{y0}a_y + H_{z0}a_z.$$

The closed line integral of H about this path is then approximately the sum of four values of H.dl he direction as on each side.

Let us choose the direction as 12341, which corresponds to a current in the a_z- direction and the first contribution is,

$$\therefore \quad (H\cdot \Delta L)_{1-2} = H_{y,1-2}\Delta y$$

∴ The value of H_y on this section of the path may be given in terms of the reference value H_{y_0} at the center of the rectangle, the rate of change of H_y with x and the distance is $\frac{\Delta x}{2}$ from the center to the midpoint of side 1-2.

$$\therefore \quad H_{y,1-2} = H_{y0} + \frac{dH_y}{dx}\left(\frac{\Delta x}{2}\right).$$

$$\Rightarrow \quad (H\cdot \Delta L)_{1-2} = H_{y0} + \left(\frac{dH_y}{dx}\left(\frac{\Delta x}{2}\right)\right)\Delta y \quad \text{-----------1}$$

Similarly along the next section of the path, we have

$$(H\cdot \Delta L)_{2-3} = H_{y,2-3}(-\Delta x) = -H_{y,2-3}\Delta x.$$

$$\Rightarrow \quad (H\cdot \Delta L)_{2-3} = -(H_{x0} + \frac{dH_x}{dy}\left(\frac{\Delta y}{2}\right))\Delta x \quad \text{-----------2}$$

Similarly for path 3-4, we have

$$(H\cdot \Delta L)_{3\text{-}4} = -\,H_y \cdot \Delta y.$$

$\Rightarrow$ $(H\cdot \Delta L)_{3\text{-}4} = -[\,H_{y0} - \frac{\Delta x}{2} \cdot \frac{dH_y}{dx}\,].\,\Delta y$ -----------3 ($\because$ $\frac{\Delta x}{2}$ is negative direction of x).

Similarly, for 4-1

$$(H\cdot \Delta L)_{4\text{-}1} = -\,H_x \cdot \Delta x$$

$\Rightarrow$ $(H\cdot \Delta L)_{4\text{-}1} = [H_{x0} - \frac{\Delta y}{2} \cdot \frac{dH_x}{dy}]\Delta x$ -----------4

Adding equations 1, 2, 3 and 4, we get

$$\oint H \cdot \Delta L = H_{y0} \cdot \Delta y + \frac{dH_y}{dx}\,\frac{\Delta x\cdot\Delta y}{2} - H_{x0}.\Delta x - \frac{dH_x}{dy}\,\frac{\Delta x\cdot\Delta y}{2}$$

$$- H_{y0} \cdot \Delta y + \frac{\Delta x\cdot\Delta y}{2} \cdot \frac{dH_y}{dx} + H_{x0} \cdot \Delta x - \frac{\Delta x\cdot\Delta y}{2} \cdot \frac{dH_x}{dy}$$

$\Rightarrow$ $\oint H \cdot \Delta L = \Delta x\cdot \Delta y\,[\frac{dH_y}{dx} - \frac{dH_x}{dy}]$

If we assume a general current density J, the enclosed current is then

$$\Delta I = J_Z \Delta x\cdot \Delta y$$

And applying Ampere's Circuital Law,

$$\oint H \cdot \Delta L = \Delta x\cdot \Delta y\,[\frac{dH_y}{dx} - \frac{dH_x}{dy}] = J_Z \Delta x\cdot \Delta y.$$

$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}} \frac{\oint H\cdot\Delta L}{\Delta x\cdot\Delta y} = \frac{dH_y}{dx} - \frac{dH_x}{dy} = J_Z$ -----------5

Similarly we can derive for J_x and J_y.

i.e. $\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}} \frac{\oint H\cdot\Delta L}{\Delta y\cdot\Delta z} = \frac{dH_z}{dy} - \frac{dH_y}{dz} = J_x$ -----------6

$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}} \frac{\oint H\cdot\Delta L}{\Delta x\cdot\Delta z} = \frac{dH_x}{dz} - \frac{dH_z}{dx} = J_y$ -----------7

$$J = J_x a_x + J_y a_y + J_z a_z.$$

$$J = \left[\frac{dH_z}{dy} - \frac{dH_y}{dz}\right] a_x + \left[\frac{dH_x}{dz} - \frac{dH_z}{dx}\right] a_y + \left[\frac{dH_y}{dx} - \frac{dH_x}{dy}\right] a_z.$$

$$J = \nabla \times H.$$

OR $$\nabla \times H = J = \lim_{\Delta SN \to 0} \int \frac{H.dl}{\Delta SN}$$

Similarly for cylindrical co-ordinates,

$$\nabla \times H = \left[\frac{1}{\rho}\frac{\partial H_z}{\partial \varphi} - \frac{\partial H_\varphi}{\partial z}\right] a_\rho + \left[\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho}\right] a_\varphi + \left[\frac{1}{\rho}\frac{\partial(\rho.H_\varphi)}{\partial \rho} - \frac{1}{\rho}\frac{\partial H_\rho}{\partial \varphi}\right] a_z.$$

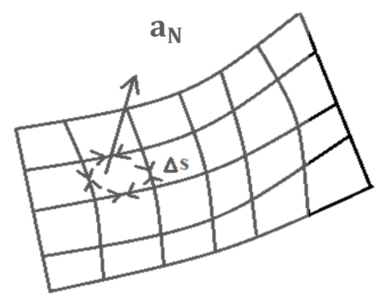
For spherical co-ordinates, we have

$$\nabla \times H = \frac{1}{r \sin\theta}\left(\frac{\partial H_\varphi \sin\theta}{\partial \theta} - \frac{\partial H_\theta}{\partial \varphi}\right) a_r + \frac{1}{r}\left(\frac{1}{\sin\theta}\frac{\partial H_r}{\partial \varphi} - \frac{\partial(r.H_\varphi)}{\partial r}\right) a_\theta + \frac{1}{r}\left(\frac{\partial(r.H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta}\right) a_\varphi.$$

Stoke's Theorem:

A line integral of a vector H around a closed a path L is equal to the integral of curl H over the open surface S enclosed by the closed path L.

The theorem is valid only when H and $\nabla \times H$ are continuous on the surface S.



Consider the surface S as shown in the figure where it is further broken into small surface area as ΔS. Applying curl definition for ΔS, we get

$$\oint \frac{H \cdot dl\Delta s}{\Delta s} = (\nabla \times H)_N$$

N ⇨ Normal to the surface.

dl · Δs ⇨ Small incremental surface as shown in figure.

⇨ $\oint H \cdot dl\Delta s = (\nabla \times H) \cdot a_N \Delta S$

⇨ $\oint H \cdot dl\Delta s = (\nabla \times H) \cdot \Delta S.$

From the figure we see that all the inner surface get cancelled due to the boundary traced in two direction.

∴ Remaining is outer surface i.e. total surface.

⇨ $\oint H \cdot dl = \oint_s (\nabla \times H) \cdot ds$ This is the Stoke's theorem.

Prove that divergence of a curl is zero using Stoke's theorem:

i.e. "$\nabla \cdot \nabla \times A = 0$"

Let A represent only vector field, then let us consider the divergence of curl vector A as scalar say T which is represented by,

$$\nabla \cdot \nabla \times A = T. \qquad \text{-----------1}$$

Integrate the above expression with respect to volume, dv

$$\Rightarrow \quad \oint_v \nabla \cdot \nabla \times A \cdot dv = \oint_v T\, dv \qquad \text{-----------2}$$

But from divergence theorem, we get

$$\oint_v \nabla \cdot (\nabla \times A) \cdot dv = \oint_s \nabla \times A\, ds$$

Then equation 2 becomes,

$$\oint_s \nabla \times A \cdot ds = \oint_v T\, dv$$

The left side is the surface integral of the curl of A over the closed surface surrounding the volume, then from Stoke's theorem, which gives

$$\oint_v T\, dv = 0.$$

Since the volume is not zero ($v \neq 0$)

$$\Rightarrow \quad T = 0$$

∴ From equation 1 it is proved that,

$$\nabla \cdot \nabla \times A = T = 0.$$

From Stokes theorem

$$\Rightarrow \quad \nabla \times H = J$$

$$\Rightarrow \quad \nabla \cdot (\nabla \times H) = \nabla \cdot J = 0.$$

$$\Rightarrow \quad \nabla \cdot J = 0.$$

Expressions for Curl in different co-ordinate system:

1. In Rectangular or Cartesian co-ordinates:

$$\nabla \times H = \begin{vmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$

2. In Cylindrical co-ordinates system:

$$\nabla \times H = \frac{1}{\rho}\begin{vmatrix} a_\rho & \rho a_\varphi & a_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \varphi} & \frac{\partial}{\partial z} \\ H_\rho & \rho\, H_\varphi & H_z \end{vmatrix} = \begin{vmatrix} \frac{a_\rho}{\rho} & a_\varphi & \frac{a_z}{\rho} \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \varphi} & \frac{\partial}{\partial z} \\ H_\rho & \rho H_\varphi & H_Z \end{vmatrix}$$

3. In Spherical co-ordinates system:

$$\nabla \times H = \frac{1}{r^2 \sin\theta}\begin{vmatrix} a_r & r a_\theta & r\sin\theta\, a_\varphi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ H_r & r\, H_\theta & r\sin\theta\;\, H_\varphi \end{vmatrix} = \begin{vmatrix} \frac{a_r}{r^2\sin\theta} & \frac{a_\theta}{r\sin\theta} & \frac{a_\varphi}{r} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ H_r & r\, H_\theta & r\sin\theta\;\, H_\varphi \end{vmatrix}$$

MODULE-4

MAGNETIC FORCES, MATERIALS

- If the current distribution is known, then we can easily determine, H, B and A at ever point in space.
- In this unit, we are going to determine the forces and torque exerted by the magnetic field on the other charges.
- The electric filed causes a force to be exerted on a charge which may be either stationary or in motion.
- We are going to study that steady magnetic field is capable of exerting a force only on a moving charges.
- A magnetic field may be produced by moving charges and may exert forces on moving charge; a magnetic field cannot arise from stationary charges and cannot exert any force on a stationary charge.
- The problems associated with the motion of particles in a vacuum are largely avoided.

FORCE ON A MOVING CHARGE.

From the electric field intensity we have

$$E = \frac{F}{Q}$$

Or force on charge particle is given as

F = QE -- 1

The force is in the same direction as the electric field intensity (for a positive charge) and is directly proportional to both E and Q.

If the charge is in motion then force at any point in its trajectory is given by equation 1. i.e

F = QE -- 1

A charge particle in motion in a magnetic field of flux density B; experience a force whose magnitude is proportional to the product of the magnitude of the charge, its velocity v, and the flux density B and sine of the angle between the vector v and B. the direction of the force is perpendicular to both v and B. and is given by a unit vector in the direction of v x B is given as

$$F = Q\, v\, B\sin\theta$$

∴ $F = Q\,(v \times B)$-- 2

The fore on a moving particle is arising from both electric field and magnetic field and the expression is obtained by superposition theorem and is given by adding equation 1 and equation 2

$F = QE + Q\,(v \times B)$

∴ **$F = Q\,(E + v \times B)$**------------------------------------ 3

The above equation 3 is called **LORENTZ FORCE EQUATION.**

LORENTZ FORCE EQUATION and its solution has some applications such as

1. It is used in determine electron orbit in the magnetron
2. Proton path in the cyclotron
3. Plasma characteristics in a magneto hydrodynamics (MHD).

FORCE ON DIFFERENTIAL CURRENT ELEMENT.

The conventional current density J can be represented in term of the velocity of the volume charge density and is given by

$J = \rho_v\, v$-------------------------------- 1

The force exerted on a differential element of charge dQ moving in a steady magnetic field is given by

$dF = dQ\,(v \times B)$---------------------- 2

But we have,

$\rho_v = \frac{dQ}{dv}$

- $dQ = \rho_v \cdot dv$ -------------------------- 3

Substitute equation 3 in equation 2 we get

$dF = \rho_v \cdot dv\,(v \times B)$ ----------------- 4

but from equation 1, above equation can be simplified as

$dF = J \times B \cdot dv$----------------------- 5

the relationship between current element is given by

$J\,dv =$ k ds = I dl ------------------ 6

∴ Lorentz force equation applied to surface current density is given by

dF = K x B $\cdot\, ds$ ---------------------- 7

∴ Lorentz force equation applied to differential current element is given by

dF = Idl x B -------------------------- 8

Integrating the equations 5, equation 7 and equation 8 we get

$$F = \oint_v J \times B \cdot dv$$

$$= \oint_s K \times B \cdot ds$$

$$F = \oint Idl \times B \quad \text{or} \quad -\oint IB \times dl$$

If the conductor is straight and the field B is uniform along it, then

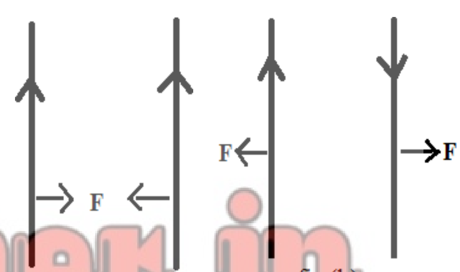
$$F = Idl \times B = ILB\sin\theta$$

Where θ is the angle between the vector representing the direction of the current flow and the direction of the magnetic flux density.

FORCE BETWEEN DIFERENTIAL CURRENT ELEMENTS:

The concept of the magnetic field was introduced to break into two parts, the problem of finding the interaction of one current (element) distribution on a second current distribution.

Let us consider two current carrying conductors are placed parallel to each other, such that each conductor produces its own flux around it, due to which there exists a force of attraction or repulsion depending upon the direction of current on the conductor.



fig(a) fig (b)

- If the direction of the both the currents are same, then the conductors experience a force attraction as shown in the figure (a).
- If the direction of the both the currents are opposite to each other, then the conductors experience force repulsion as shown in the figure (b).

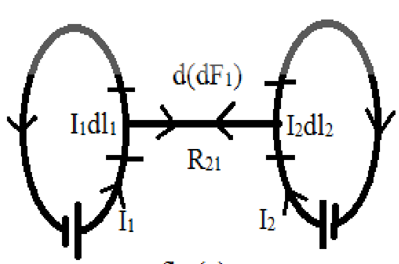


fig (c)

- Let us consider two current elements I_1dl_1 and I_2dl_2 as shown in the figure (c).
- Note the direction of current I_1 and I_2 are same.
- Since the current is in the same direction, the force of attraction is as shown in figure (c).

From the equation of force, the force exerted on a differential current element is given by,

$$d(dF_1) = I_1 dl_1 \times dB_2 \text{ --- } 1$$

But from BIOT-SAVART LAW

$$dB_2 = \mu_0 dH_2 = \mu_0 \frac{I_2 dl_2 \times a_{R_{21}}}{4\pi R_{21}^2} \text{ ------}2$$

Substituting equation (2) in equation (1), we get

$$d(dF_1) = \mu_0 \frac{I_1 dl_1 \times (I_2 dl_2 \times a_{R_{21}})}{4\pi R_{21}^2}$$

Integrating above equation twice, we get

$$F_1 = \frac{\mu_0 I_1 I_2}{4\pi} \oint_{L_1} \oint_{L_2} \frac{dl_1 \times (dl_2 \times a_{R_{21}})}{R_{21}^2} \text{ ------}3$$

Similarly we get

$$F_2 = \frac{\mu_0 I_1 I_2}{4\pi} \oint_{L_1} \oint_{L_2} \frac{dl_2 \times (dl_1 \times a_{R_{12}})}{R_{12}^2} \text{ ------}4$$

From the above equations (3) and (4) are opposite to each other

i.e. $F_1 = -F_2$ or $F_2 = -F_1$ --5

Thus from above we see that it obeys Newton's third law, i.e. for every action there is equal and opposite reaction.

∴ For the two current carrying conductors of length L each, the force exerted is given by

$$\mathbf{F} = \frac{\mu I_1 I_2 L}{2\pi d}$$

Where I_1 and I_2 are the current flowing through conductor 1 and conductor 2 and d is the distance of separation between two conductors.

MAGNETIC BOUNDARY CONDITIONS:

The conditions of the magnetic field existing at the boundary of the two media when the magnetic field passes from one medium to other are called boundary conditions.

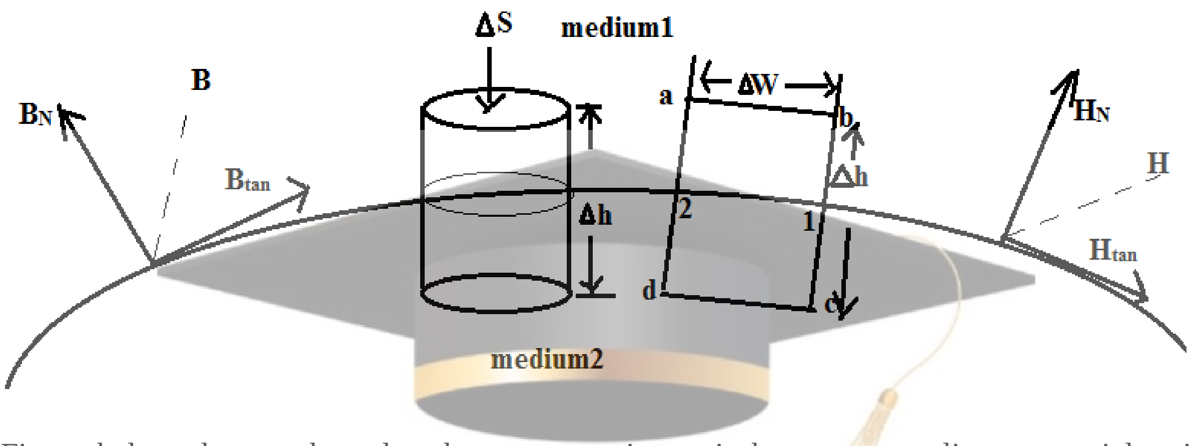


Figure below shows a boundary between two isotropic homogeneous linear materials with permeability μ_1 and μ_2.

To study the boundary conditions let us consider two cases.

1. Normal to the boundary.
2. Tangential to the boundary.

Boundary conditions for normal component:

To find the normal component of B, let us select a closed Gaussian surface in the form of a right circular cylinder as shown in figure.

Let the height of the cylinder be Δh and be placed in such a way that $\frac{\Delta h}{2}$ is in medium1 and remaining $\frac{\Delta h}{2}$ is in medium2, and axis of the cylinder is in the normal direction to the surface.

But Gauss's law for magnetic field is,

$$\oint_s B \cdot ds = 0 \text{ ---1}$$

⇨ L.H.S. is given by

$$\oint_s B \cdot ds = \oint_{top} B \cdot ds + \oint_{bottom} B \cdot ds + \oint_{lateral} B \cdot ds = 0 \text{ ---------------2}$$

To calculate the boundary conditions let $\Delta h = 0$ then surface integral is calculated for top and bottom surface only.

Since these surfaces are very small. Let the magnitude of normal component of B be B_{N1} and B_{N2} in medium1 and medium2 respectively.

$$\therefore \quad \oint_{top} B \cdot ds = B_{N1} \oint_{top} ds = B_{N1} \cdot \Delta s \text{ ------------------------------3}$$

Similarly, $\oint_{bottom} B\cdot ds = B_{N2} \oint_{bottom} ds = B_{N2}\cdot \Delta s$ ----------------------4

$\oint_{lateral} B\cdot ds = 0 \quad \Rightarrow \quad \Delta h = 0$

But from figure, normal component is entering medium1 and leaving the medium2 so B_{N1} and B_{N2} are opposite to each other. Then equation2 becomes,

$\therefore \quad B_{N1}\cdot \Delta s - B_{N2}\cdot \Delta s = 0$

$\Rightarrow \quad \cancel{\Delta s}\, B_{N1} = \cancel{\Delta s}\, B_{N2}$

$\Rightarrow \quad B_{N1} = B_{N2}$ --5

Thus the normal component of B is continuous at boundary.

But $B = \mu H$ $\Rightarrow$ equation (5) can be written as,

$\Rightarrow \quad \mu_1 H_{N1} = \mu_2 H_{N2}$

$\Rightarrow \quad \frac{H_{N1}}{H_{N2}} = \frac{\mu_2}{\mu_1}$

$\therefore \quad \boldsymbol{\frac{H_{N1}}{H_{N2}} = \frac{\mu_2}{\mu_1} = \frac{\mu_{r2}}{\mu_{r1}}}$

$\therefore$ Normal component of H is not continuous at the boundary.

i.e. Field strength in two media are inversely proportional to their relative permeability.

Boundary conditions for tangential component:

From Ampere's circuital law we have,

$\oint H\cdot dl = I$ --1

Consider rectangular path a-b-c-d-a as shown in the figure, which is placed in a plane normal to the boundary surface, then equation(1) can be written as,

$$\oint H\cdot dl = \oint_a^b H\cdot dl + \oint_b^1 H\cdot dl \oint_1^c H\cdot dl + \oint_c^d H\cdot dl \oint_d^2 H\cdot dl + \oint_2^a H\cdot dl = I \text{ --------------2}$$

From the figure it is clear that, closed path is placed in such a way that its two sides a-b and c-d are parallel to the tangential direction to the surface while the other two sides are normal to the surface at the boundary.

- This closed path is placed in such a way that half of its portion is in medium1 and remaining is in medium2.

From the figure it is clear that the normal and the tangential components in medium1 and medim2 are in opposite direction and equation(2) can be written as,

$$K \cdot dw = H_{tan1}(\Delta w) + H_{N1}\left(\frac{\Delta h}{2}\right) + H_{N2}\left(\frac{\Delta h}{2}\right) - H_{tan2}(\Delta w) - H_{N2}\left(\frac{\Delta h}{2}\right) - H_{N1}\left(\frac{\Delta h}{2}\right) \text{------------------} 3$$

At boundary condition $\Delta h = 0$

⇨ $K \cdot dw = H_{tan1}(\Delta w) - H_{tan2}(\Delta w)$

⇨ $K = H_{tan1} - H_{tan2}$ --- 4

In vector form above equation can be expressed as,

$H_{tan1} - H_{tan2} = a_{N_{12}} \times K$ --5

Where $a_{N_{12}}$ is the unit vector in the direction normal at the boundary from medium1 to medium2, where K ⇨ surface current density.

But $B = \mu H$

⇨ $\frac{B_{tan1}}{\mu 1} - \frac{B_{tan2}}{\mu 2} = K$

- Consider a special case with current at the boundary, then

$H_{tan1} = H_{tan2}$ --- 6

OR $\frac{B_{tan1}}{\mu 1} = \frac{B_{tan2}}{\mu 2}$

$\frac{B_{tan1}}{B_{tan2}} - \frac{\mu_1}{\mu_2}$ ------7

From equation (6) the tangential component of H are continuous in nature.

And from equation (7) B are discontinuous at the boundary, with the condition that the boundary is current free.

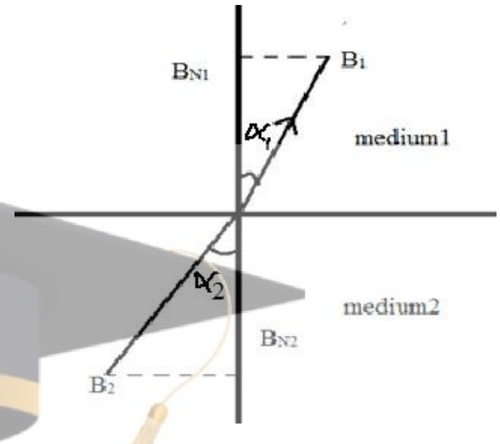
Let us consider the field makes an angle α_1 and α_2 with normal to the interface as shown in the figure. Then,

In medium1,

$\tan \alpha_1 = \frac{B_{tan1}}{B_{N1}}$ ------8

In medium2,

$$\tan\alpha_2 = \frac{B_{tan2}}{B_{N2}} \qquad \text{----- 9}$$



Dividing equation (8) and equation (9), we get

$$\frac{\tan\boldsymbol{\alpha}_1}{\tan\boldsymbol{\alpha}_2} = \frac{B_{tan1}}{\cancel{B_{N1}}} \cdot \frac{\cancel{B_{N2}}}{B_{tan2}}$$

Since $B_{N1} = B_{N2}$

⇨ $$\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{B_{tan1}}{B_{tan2}} = \frac{\mu_{r1}}{\mu_{r2}}$$

MAGNETIC CIRCUITS:

In magnetic circuits, we determine the magnetic fluxes and magnetic field intensities in various parts of the circuits. If we study the analogy between the electric and magnetic circuits, then it becomes very easy for us to understand the magnetic circuit.

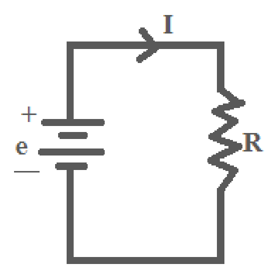


fig (a)

Let us study analogy between the electric and magnetic circuits. An electric circuit with closed path form a circuit similar to these magnetic lines of flux is continuous and can form closed path.

So a single magnetic line of flux or all parallel magnetic lines of flux may be considered as magnetic circuit.

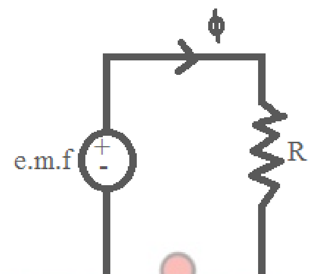


fig (b)

Similar to electromotive force (e.m.f) in an electronic circuit, we can define a new quantity in case of a magnetic circuit called magneto motive force (m.m.f). The magneto motive force is defined as,

$$e_m = NI = \int H \cdot dl$$

S. I. unit of m.m.f is Ampere (A).

Generally in magnetic circuits, the source of m.m.f is a coil carrying conductors with N number of turns as shown in the figure (c).

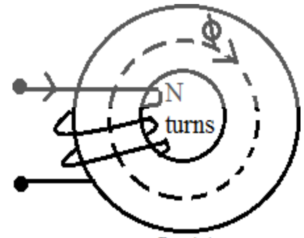


fig (c)

∴ m.m.f is measured in Ampere-turn (A-t).

In electronic circuits we have R = $\frac{V}{I}$

Similarly analogically in magnetic field reluctance is the ratio of the magneto motive force to the total flux.

$R = \frac{e_m}{\varphi}$

⇨ reluctance is measured in $\frac{\text{Ampere−turn}}{\text{weber}}$

The resistance in electric circuit can be expressed in terms of conductivity σ as,

$$R = \frac{1}{\sigma s}$$

—

l ⇨ length, s ⇨ cross section, σ ⇨ conductivity of the linear isotropic homogeneous material. In case of magnetic circuits, we can define reluctance in very much similar way as,

$$R = \frac{1}{\mu s}$$

where μ ⇨ permeability.

MODULE-5

TIME VARYING FIELDS AND MAXWELLS EQUATIONS

In this unit, we will introduce two new concepts, i.e

- The electric field is produced by changing magnetic field and
- The magnetic field is produced by changing electric field.

These topics were researched by a person called Michel Faraday, and it was proved by Maxwell.

FARADAY'S LAW:-

Faraday introduces the concept i.e. a current could produce a magnetic field, then the magnetic field should be able to produce a current.

We now say that time varying magnetic field produces an electromotive force (e.m.f), which may establish a current in a suitable closed path.

This e.m.f is merely a voltage that arises from conductor moving in a magnetic field or from changing magnetic field.

Faraday discovered that the induced e.m.f is equal to the rate of change of flux (magnetic flux).

i.e e.m.f = $-\frac{\partial \varphi}{\partial t}$ Volts -------1

A non-zero value of may result from any of the following situation.

* A time changing flux linking a stationary closed path.
* Relative motion between a steady flux and a closed path.
* A combination of two.

The minus sign is an indication that the e.m.f. induced is in such a direction that produces a current whose flux if added to the original flux, would reduce the magnitude of the e.m.f.

This statement that the induced voltage acts to produce an opposing flux is known as **LENZ'S LAW.**

If the closed path is that taken by a N-turn filamentary conductor then e.m.f is

e.m.f = $-N\frac{\partial \varphi}{\partial t}$ -------2

where φ is now interpreted as the flux passing through any one of N co-incident path. Since the induced e.m.f is obviously a scalar, and is measured in volts then we define e.m.f = $\int$ E . dl -------3

And note that it is the voltage about a specific closed path.

If any part of the path is changed, e.m.f also changes.

But we have B = $\frac{\partial \varphi}{\partial s}$

⇨ $\partial\varphi$ = B . ds

⇨ $\varphi = \int$ B . ds -------4

From equation 1

e.m.f = - $\frac{\partial \varphi}{\partial t}$

∴ e.m.f = - $\frac{\partial}{\partial t}$ ($\int$ B . ds) = $\int$ E . dl -------5

Where the fingers of our right hand indicate the direction of the closed path, and our thumb indicates the direction of ds.

A flux density B in the direction of ds and increasing with time thus produces an average value of E which is opposite to the positive direction about the closed path.

Let us divide our studies into parts by first finding the combination to the total e.m.f made by changing field with in

(i) Stationary path (transformer e.m.f) and
(ii) Moving path within a constant(motional or generator or e.m.f)

(i) Stationary Path:- From equation 5 we have

e.m.f = - $\frac{\partial}{\partial t}$ ($\int$ B . ds) = $\int$ E . dl

In the above expression, the magnetic flux is the only time-varying quantity on the right hand side then the partial derivative may be taken under the integral sign.

∴ e.m.f = $\int$ E . dl = - $\int \frac{\partial B}{\partial t}$ ds -------6

But from stokes theorem we can write

$\int$ E . dl = $\int$(∇ X E) ds -------7

Equating equation 6 and 7

$\int$(∇ X E) ds = - $\int \frac{\partial B}{\partial t}$ ds

⇨ $\nabla \times E = -\frac{\partial B}{\partial t}$

The above equation is called Maxwell's equation in differential form or point form.

From equation 6 we have

$$\int E \,.\, dl = -\int \frac{\partial B}{\partial t}\, ds$$

The above equation is Maxwell's equation in Integral form.

If B is not a function of time then

$\frac{\partial B}{\partial t} = 0$

⇨ $\int E \,.\, dl = 0$ or $\nabla \times E = 0$ these are called Electro static

(ii). Second only we consider moving path with constant (motional e.m.f)

Let the force on charge on Q moving at a velocity v, in a magnetic field B is given by

$F = Q.\, v \times B$

$\frac{F}{Q} = v \times B$

⇨ $E_m = v \times B$ motional electric field intensity.

But e.m.f = $\int E \,.\, dl$

⇨ e.m.f = $\int v \times B \,.\, dl$

where above equation is a non-zero value only along that portion of the path which is in motion.

If we consider the magnetic flux density is also varying with time then

e.m.f = $\int E \,.\, dl = = -\int \frac{\partial B}{\partial t}\, ds + \int v \times B \,.dl$

Displacement Current:-

From Faraday's law we got the Maxwell's equation given by

$\nabla \times E = -\frac{\partial B}{\partial t}$ ---------1

From the above equation 1 we can analyze that, time varying magnetic field produces an electric field,

from ACL, we have steady magnetic field in point form is given by

$\nabla \times H = J$ ---------2

Taking divergence on both sides of the equation 2, we get

$\nabla.(\nabla \text{ x H}) = \nabla.\text{J}$

But from CURL property we have

$\nabla.(\nabla \text{ x H}) = 0$

Then we get

$\nabla.\text{J} = 0$ ----------3

We also have the equation for continuity in terms of current density given by

$\nabla.\text{J} = -\frac{\partial \rho_v}{\partial t}$ ------------4

If $\frac{\partial \rho_v}{\partial t} = 0$ then only it satisfies

$\nabla.\text{J} = 0$

⇨ $\nabla.(\nabla \text{ x H}) = 0$

Which is unrealistic or not compatible with time varying field?

⇨ We have to add some unknown term G in equation 2 i.e $\nabla \text{ x H} = \text{J}$

⇨ $\nabla \text{ x H} = \text{J} + \text{G}$ ------------5

Now again taking divergence both sides for equation 5 we

$\nabla.(\nabla \text{ x H}) = \nabla.\text{J} + \nabla.\text{G}$

Applying curl property to above equation we get

$\nabla.(\nabla \text{ x H}) = \nabla.\text{J} + \nabla.\text{G} = 0$

$\nabla.\text{J} + \nabla.\text{G} = 0$

but we have

$\nabla.\text{J} = -\frac{\partial \rho_v}{\partial t}$

$-\frac{\partial \rho_v}{\partial t} + \nabla.\text{G} = 0$

⇨ $\nabla.\text{G} = \frac{\partial \rho_v}{\partial t}$

but $\rho_v = \nabla . D$

⇨ $\nabla.\text{G} = \frac{\partial(\nabla.D)}{\partial t}$

Interchanging the $\frac{\partial}{\partial t}$ and ∇

We get $\nabla.\text{G} = \nabla.\frac{\partial D}{\partial t}$

Comparing L.H.S and R.H.S we get

$$G = \frac{\partial D}{\partial t} \qquad \text{------------6}$$

Therefore equation 5 becomes

$$\nabla \times H = J + \frac{\partial D}{\partial t} \qquad \text{---------7}$$

Therefore the above equation is the second Maxwell's Equation, and the additional term $\frac{\partial D}{\partial t}$ has the dimension of current density A/m^2. And since it varies w.r.t to time for electric field, So Maxwell named it as displacement current. i.e

$$J_D = \frac{\partial D}{\partial t}$$

- ⇨ Equation 7 $\nabla \times H = J + J_D$

  Were $J = \sigma E$ is called conduction current density

  and $J_D = \frac{\partial D}{\partial t}$ is called Displacement Current Density.

Now, if we consider the non-conducting medium in which volume charge density is zero then

- ⇨ $J = 0$

Then equation 7 becomes

$$\nabla \times H = \frac{\partial D}{\partial t} \qquad \text{-------8}$$

But we have b

$$\nabla \times E = -\frac{\partial B}{\partial t} \qquad \text{------9}$$

Equation 8 and 9 are symmetry to each other

Therefore $J_D = \frac{\partial D}{\partial t}$ is called point form of Displacement Current.

Taking integration for the above equation of J_D w.r.t to surface we get

$$I_D = \oint J_D \, ds = \oint \frac{\partial D}{\partial t} \, ds$$

Therefore taking integral w.r.t surface for equation 7 we get

$$\oint (\nabla \times H) ds = \oint J \, ds + \oint \frac{\partial D}{\partial t} \, ds$$

But from stroke's theorem we get

$$\oint H.dl = I + \oint \frac{\partial D}{\partial t} \, ds$$ this equation is the Maxwell's equation in Integral form.

Maxwell's Equation in Point Form or Differential Form

1. $\nabla \times E = -\frac{\partial B}{\partial t}$ From Faraday's Law
2. $\nabla \times H = J + \frac{\partial D}{\partial t}$ From Curl
3. $\nabla . D = \rho_v$ From Gauss's Law
4. $\nabla . B = 0$ From Biot-Savart Law

Maxwell's Equation in Integral form

1. $\oint E.dl = -\oint \frac{\partial B}{\partial t} ds$ From Faraday's Law
2. $\oint H.dl = I + \oint \frac{\partial D}{\partial t} ds$ From Ampere's Circuital Law
3. $\oint D\, ds = \oint \rho_v\, dv$ From Gauss's Law
4. $\oint B\, ds = 0$ From Biot-Savart Law

UNIFORM PLANE WAVES

The existence of electromagnetic (EM) wave was stated by prof. Heinrich Hertz.

Maxwell himself predicted the existence of EM waves earlier.

Hertz was the first scientist who generated and detected radio waves successfully, where waves are means of transporting energy or information from source to destination.

Uniform plane wave in free space:

Uniform plane waves are those waves, where electric and magnetic fields are uniform everywhere. There is no variation at any point.

i.e. Consider that an electric field is in x-direction, while a magnetic field in is y-direction. Both the fields will not vary with x and y directions but vary in only z-direction.

∴ Consider Maxwell's equation expressed in E and M as,

$$\nabla \times H = J + \frac{\partial D}{\partial t}$$

Let us assume that a free space is perfect dielectric, then J=0. Because $\sigma = 0$.

$$\nabla \times H = \frac{\partial D}{\partial t}$$

D is a rectangular co-ordinates system. Then

$$\left[\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right] a_x + \left[\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x}\right] a_y + \left[\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}\right] a_z = \frac{\partial}{\partial t}\left[\Delta x a_x + \Delta y a_y + \Delta z a_z\right].$$

As H is in y-direction ⇨ $H_x = H_z = 0$.

$$-\frac{\partial H_y}{\partial z} a_x + \frac{\partial H_y}{dx} a_z = \frac{\partial D}{\partial t}.$$

Since H_y is not changing with x, because wave is in z direction so we have

⇨ $\frac{\partial H_y}{dx} = 0.$

⇨ $-\frac{\partial H_y}{\partial z} a_x = \frac{\partial D_x}{\partial t}.$

But $D = \epsilon E$.

⇨ $\frac{\partial H_y}{\partial z} a_x = -\epsilon \frac{\partial}{\partial t}(E_x a_x + E_y a_y + E_z a_z).$

Equating the coefficient of the vector direction, then we get

⇨ $\frac{\partial H_y}{\partial z} = -\epsilon \frac{\partial E_x}{\partial t}$ -----------1

Again we have to consider Maxwell's equation,

$$\nabla \times E = -\frac{\partial B}{\partial t}.$$

$$\left[\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y}\right] a_x + \left[\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right] a_y + \left[\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x}\right] a_z = -\mu \frac{\partial}{\partial t} H_x a_x + H_y a_y + H_z a_z)..$$

As E is in x- direction ⇨ $E_y = E_z = 0$. And $H_x = H_z = 0$.

$$\frac{\partial E_x}{\partial z} a_y + \frac{\partial E_x}{\partial y} a_z = -\mu \frac{\partial}{\partial t} H_y a_y .$$

Since E_x is not varying with y, because wave I in z direction so above equation is written as,

⇨ $\frac{\partial E_x}{\partial y} = 0$

$\Rightarrow \quad \frac{\partial E_x}{\partial z} a_y = -\mu \frac{\partial}{\partial t} H_y a_y$.

Equating the coefficient of the above equation, we get,

$\Rightarrow \quad \frac{\partial E_x}{\partial z} = -\mu \frac{\partial H_y}{\partial t}$. ----------2

Differentiating equation 1 with respect to t, we get.

$$\frac{\partial}{\partial t}\left[\frac{\partial H_y}{\partial z}\right] = -\epsilon \frac{\partial^2 E_x}{\partial t^2} \qquad \text{----------3}$$

Differentiating equation 2 with respect to z, we have.

$$\frac{\partial}{\partial z}\left[\frac{\partial H}{\partial t}\right] = -\frac{1}{\mu}\frac{\partial^2 E_x}{\partial z^2} \qquad \text{----------4}$$

From equations 3 and 4 both LHS are same after changing the order of differentiation.

$$\epsilon \frac{\partial^2 E_x}{\partial t^2} = \frac{1}{\mu}\frac{\partial^2 E_x}{\partial z^2}.$$

$\Rightarrow \quad \frac{\partial^2 E_x}{\partial t^2} = \frac{1}{\mu\epsilon}\frac{\partial^2 E_x}{\partial z^2}.$

But we have $\quad \nu = \frac{1}{\sqrt{\mu\epsilon}}$

$\Rightarrow \quad \frac{\partial^2 E_x}{\partial t^2} = \nu^2 \frac{\partial^2 E_x}{\partial z^2}.$

where ν is the wave velocity

∴ Above equation is called wave equation.

Wave equation in free space:

Consider $\quad \nabla \times E = -\frac{\partial B}{\partial t}$.

Taking curl of both sides, we have

$$\nabla \times \nabla \times E = -\nabla \times \frac{\partial B}{\partial t} .$$

$$\nabla(\nabla \cdot \mathrm{E}) - \nabla^2 \cdot \mathrm{E} = -\frac{\partial}{\partial \mathrm{t}}.(\nabla \times \mathrm{B})$$

We know that, $\mathrm{D} = \epsilon \mathrm{E}$ and $\mathrm{B} = \mu \mathrm{H}$.

$$\frac{1}{\epsilon}[\nabla(\nabla \cdot \mathrm{D})] - [\nabla^2 \mathrm{E}] = -\frac{\partial}{\partial \mathrm{t}}(\nabla \times \mu \mathrm{H}). \qquad (\because \nabla \cdot \mathrm{D} = \rho_v)$$

$$\frac{1}{\epsilon}\nabla \cdot \rho_v - \nabla^2 \cdot \mathrm{E} = -\mu \frac{\partial}{\partial \mathrm{t}}(\nabla \times \mathrm{H}).$$

But $\nabla \times \mathrm{H} = \mathrm{J} + \frac{\partial D}{\partial t}$ [Ampere's Circuital Law].

Substituting above expression, we get

$$\frac{1}{\epsilon}\nabla \cdot \rho_v - \nabla^2 \cdot \mathrm{E} = -\mu \frac{\partial}{\partial \mathrm{t}}\left(\mathrm{J} + \frac{\partial D}{\partial t}\right).$$

In free space $\rho_v = 0$ and $\mathrm{J} = 0$.

$$\Rightarrow \quad -\nabla^2 \cdot \mathrm{E} = -\mu \frac{\partial^2 \mathrm{D}}{\partial \mathrm{t}^2}.$$

$$\Rightarrow \quad \nabla^2 \cdot \mathrm{E} - \mu \frac{\partial^2 \mathrm{D}}{\partial \mathrm{t}^2} = 0.$$

But $\mathrm{D} = \epsilon \mathrm{E}$.

$$\Rightarrow \quad \nabla^2 \cdot \mathrm{E} - \mu\epsilon \frac{\partial^2 \mathrm{E}}{\partial \mathrm{t}^2} = 0.$$

In terms of H.

Consider $\nabla \times \mathrm{H} = \mathrm{J} + \frac{\partial D}{\partial t}$

Taking curl on both sides, we get

$$\nabla \times \nabla \times \mathrm{H} = \nabla \times \mathrm{J} + \frac{\partial}{\partial t}(\nabla \times \mathrm{D}).$$

$$\nabla(\nabla \cdot \mathrm{H}) - \nabla^2 \cdot \mathrm{H} = \nabla \times \mathrm{J} + \frac{\partial}{\partial t}(\nabla \times \mathrm{D}).$$

We have $\mathrm{B} = \mu \mathrm{H} \quad \Rightarrow \quad \mathrm{H} = \frac{\mathrm{B}}{\mu}$

$\mathrm{J} = \sigma \mathrm{e} \qquad \mathrm{D} = \epsilon \mathrm{E}$

$$\frac{1}{\mu}\nabla(\nabla\cdot B) - \nabla^2 \cdot H = \sigma\,(\nabla \times E) + \epsilon\frac{\partial}{\partial t}(\nabla \times E).$$

$\nabla \cdot B = 0$ for free space $\sigma = 0$.

$$-\nabla^2 \cdot H = \epsilon\frac{\partial}{\partial t}\left(-\frac{\partial B}{\partial t}\right).$$

$$-\nabla^2 \cdot H = -\epsilon\frac{\partial^2 B}{\partial t^2}.$$

$$-\nabla^2 \cdot H = -\epsilon\mu\frac{\partial^2 H}{\partial t^2}.$$

$$\nabla^2 \cdot H - \epsilon\mu\frac{\partial^2 H}{\partial t^2} = 0.$$

Relation between E and H in uniform plane wave:

$$\nabla \times H = J + \frac{\partial D}{\partial t}$$

In free space J = 0.

⇨ $\nabla \times H = \frac{\partial D}{\partial t} = \frac{\partial(\epsilon E)}{\partial t} = \epsilon\frac{\partial E}{\partial t}.$

⇨ $\nabla \times H = \begin{vmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}.$

⇨ $\nabla \times H = a_x\left[\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right] - a_y\left[\frac{\partial H_z}{\partial x} - \frac{\partial H_x}{\partial z}\right] + a_z\left[\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}\right] = \epsilon\frac{\partial E}{\partial t}.$

If we consider the wave to be uniform plane wave travelling in x-direction, then

⇨ $-a_y\frac{\partial H_z}{\partial x} + a_z\frac{\partial H_y}{\partial x} = \epsilon\frac{\partial E_x}{\partial t} + \epsilon\frac{\partial E_y}{\partial t} + \epsilon\frac{\partial E_z}{\partial t}.$

⇨ Equating the vectors, we have

$$-\frac{\partial H_z}{\partial x} = \epsilon\frac{\partial E_y}{\partial t} \qquad \text{---------}1$$

$$\frac{\partial H_y}{\partial x} = \epsilon \frac{\partial E_z}{\partial t} \qquad \text{---------2}$$

We have

$$\nabla^2 \cdot E_y \; - \mu\epsilon \frac{\partial^2 E_y}{\partial t^2} = 0.$$

Let the solution of the second order differential equation be given by,

$$E_y = f_1(x - \nu\, t) \qquad \text{where } \nu = \frac{1}{\sqrt{\mu\epsilon}}$$

Let $\quad u = x - \nu t \quad \Rightarrow \quad du = \; -\nu \cdot dt$

$\Rightarrow \quad E_y = f_1(x)$

$$\frac{\partial E_y}{\partial t} = \frac{\partial f_1(x)}{\partial x} \cdot \frac{\partial u}{\partial t}$$

$$\frac{\partial E_y}{\partial t} = f_1{}'(x) \cdot (-\nu) = -\nu \cdot f_1{}'(x) \qquad \text{---------3}$$

Substituting above equation in (1), we get

$$\frac{\partial H_z}{\partial x} = \epsilon\nu \cdot f_1{}'(x) \qquad \text{---------4}$$

Consider $\quad E_y = f_1(u) \qquad \frac{\partial u}{\partial x} = 1$

$$\frac{\partial E_y}{\partial x} = \frac{\partial f_1(u)}{\partial u} \cdot \frac{\partial u}{\partial x}$$

$$\frac{\partial E_y}{\partial x} = f_1{}'(u) \qquad \text{---------5}$$

Substitute equation 5in 3, we have

$$\frac{\partial E_y}{\partial t} = -\frac{\partial E_y}{\partial x} \cdot \nu$$

But $\quad \nu = \frac{1}{\sqrt{\mu\epsilon}}$

$$\frac{\partial E_y}{\partial t} = -\frac{1}{\sqrt{\mu\epsilon}} \frac{\partial E_y}{\partial x} \qquad \text{----------6}$$

Put equation 6 in 1, we have

$$-\frac{\partial H_z}{\partial x} = -\frac{\epsilon}{\sqrt{\mu\epsilon}}\frac{\partial E_y}{\partial x}$$

$$\frac{\partial H_z}{\partial x} = \sqrt{\frac{\epsilon}{\mu}}\frac{\partial E_y}{\partial x}$$

Integrate with respect to x,

$$H_z = \sqrt{\frac{\epsilon}{\mu}}E_y$$

$$\frac{E_y}{H_z} = \sqrt{\frac{\mu}{\epsilon}}$$

$\eta = Z = \frac{E}{H} = \sqrt{\frac{\mu}{\epsilon}}$ where Z is called Intrinsic Impedance of the medium

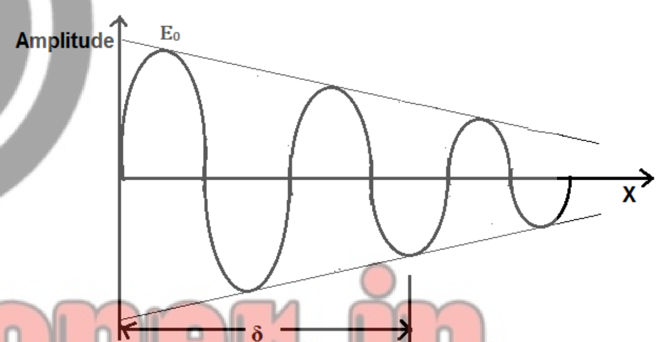
But for free space,

$$\eta = Z = \frac{E}{H} = \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{8.854 \times 10^{-12}}} = 377\Omega$$

Therefore for free space Z = 377Ω.

Skin Depth or Depth of Penetration:

It is defined as the depth of a conductor at which the amplitude of the wave has been attenuated to $\frac{1}{e}$ or 0.367 or 36.7% of the initial amplitude is called Skin depth.



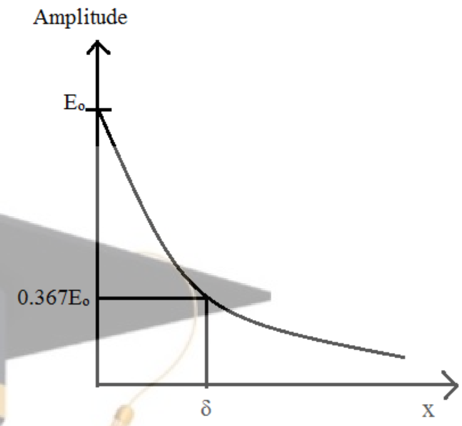
When an electromagnetic wave enters a conducting medium, its amplitude starts attenuating and finally it reaches to zero after certain distance, as a result the current induced by the wave exists only near the surface of the conductor, this effect is called a Skin depth.

If x is the distance travelled in the medium and E_o is the maximum amplitude at the initial incidence then we can write,

$$E = E_o e^{-\alpha x}$$

At $x = \delta = \frac{1}{\alpha}$ where δ is called Skin depth or depth of penetration.

Then expression of E becomes,



$$E = E_o e^{-1}$$

$$E = \frac{E_o}{e}$$

$$E = 0.367\ E_o \text{ or } 36.7\% \text{ of } E_o$$

For good conductor,

We have

$$\alpha = \sqrt{\frac{\omega\mu\sigma}{2}}$$

then

$$\delta = \frac{1}{\alpha}$$

$$\delta = \frac{1}{\sqrt{\frac{\omega\mu\sigma}{2}}} = \sqrt{\frac{2}{\omega\mu\sigma}} = \sqrt{\frac{2}{2\pi f\mu\sigma}} = \sqrt{\frac{1}{\pi f\mu\sigma}}$$

$$\delta = \sqrt{\frac{1}{\pi f\mu\sigma}}$$

Poynting theorem

Poynting theorem is the mathematical expression of the law of conservation of energy as applied to the electromagnetic fields.

Statement:- At any point in an electromagnetic field the power per unit area or power density vector is given by

$$S = E \times H \qquad \frac{\text{watt}}{\text{m}^2}$$

The vector S is given the name Poynting vector. The direction is perpendicular to the plane E and H.

To derive the expression we have from Maxwell's equation,

$$\nabla \times H = J + \frac{\partial D}{\partial t}$$

Multiply above equation with E both side we get,

$$E \cdot (\nabla \times H) = E \cdot \left(J + \frac{\partial D}{\partial t}\right) \quad \text{--------1}$$

But from vector identity,

$$\nabla(E \times H) = H \cdot (\nabla \times E) - E \cdot (\nabla \times H)$$

$$\Rightarrow \quad E \cdot (\nabla \times H) = H \cdot (\nabla \times E) - \nabla(E \times H)$$

Substitute in the above equation in equation 1we get,

$$H \cdot (\nabla \times E) - \nabla(E \times H) = E \cdot J + E \cdot \frac{\partial D}{\partial t}$$

But $J = \sigma E \qquad D = \epsilon E$

$$-\nabla(E \times H) = E \cdot J + E \cdot \frac{\partial D}{\partial t} - H \cdot (\nabla \times E)$$

But $\nabla \times E = -\frac{dB}{dt}$

$$-\nabla(E \times H) = E \cdot J + E \cdot \frac{\partial D}{\partial t} - H \cdot \left(-\frac{dB}{dt}\right)$$

$$-\nabla(E \times H) = E \cdot J + E \cdot \frac{\partial}{\partial t}(\epsilon E) + H \cdot \left(\frac{d}{dt}(\mu H)\right)$$

$$-\nabla(E \times H) = E \cdot J + \epsilon E \cdot \frac{\partial}{\partial t}(E) + \mu H \cdot \left(\frac{d}{dt}(H)\right) \quad \text{--------2}$$

But $\qquad \epsilon E \cdot \frac{\partial}{\partial t}(E) = \frac{\partial}{\partial t}\left(\frac{\epsilon E^2}{2}\right)$

$$\mu H \cdot \left(\frac{d}{dt}(H)\right) = \frac{\partial}{\partial t}\left(\frac{\epsilon H^2}{2}\right)$$

Therefore above equation 2 simplifies to,

$$-\nabla(E \times H) = E \cdot J + \frac{\partial}{\partial t}\left(\frac{\epsilon E^2}{2}\right) + \frac{\partial}{\partial t}\left(\frac{\epsilon H^2}{2}\right) \qquad \text{-------3}$$

The above equation is the point form or differential form of the Poynting theorem.

Integrating the equation 3 with respect to volume, we get

$$-\int \nabla(E \times H) \cdot dv = \int (E \cdot J) \cdot dv + \int \frac{\partial}{\partial t}\left(\frac{\epsilon E^2}{2}\right) \cdot dv + \int \frac{\partial}{\partial t}\left(\frac{\epsilon H^2}{2}\right) \cdot dv$$